Nimbus: Mathematics, theory and physical assumptions

Nimbus project documentation

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Chapter 1

Model, variables and assumptions

Nimbus represents an atmosphere with resolved motion, bulk water species and parameterized unresolved processes. Its solver has been independently developed from the beginning; it is not a port or fork of CM1. Shared scientific references and supported CM1 data conventions do not imply an identical discretization or implementation. This guide explains the mathematical choices in the implementation, why they matter, and where their limits are. It is a guide to the current source and permanent tests, not a claim that every available scheme combination is ready or that a simulation has been independently scientifically reviewed.

The executable's `--scheme-catalog=json` output describes the admitted scopes and remaining blockers. The underlying authority is the readiness registry, together with the strict configuration and runtime guards. A paper supplies a physical or numerical model; Nimbus's storage conventions, coupling order and supported configurations are additional implementation choices. Those distinctions are essential when comparing results with a paper or another atmospheric model.

The companion chapters develop the reasoning in more detail: grid, resolved dynamics and the coupled timestep derives the inventory and staggered-grid relationships and follows a complete source/pressure sequence; bulk microphysics and thermal coupling explains each scheme family's process and spectral assumptions, then separates ordinary thermal sources from the guarded source/EOS work.

1.1 What the state variables mean

The stored density `rho` is dry-air density, denoted by ρ_d. Each water variable is mass of that species divided by dry-air mass:

$$
\begin{aligned}
q_x &= \frac{m_x}{m_d}, \\
q_t &= q_v + q_c + q_r + q_i + q_s + q_g + q_h, \\
\rho_m &= \rho_d(1 + q_t).
\end{aligned}
$$

The subscripts denote vapor, cloud liquid, rain, cloud ice, snow, graupel and hail. Thus `qv = 0.01` means 0.01 kg of vapor per kg of dry air. It is not the vapor mass fraction of the complete moist mixture. Confusing these definitions changes water inventories, pressure relationships and source units. A scalar cell of physical volume V contains dry mass $\rho_d V$ and water mass $\rho_d V q_t$.

Nimbus stores potential temperature θ and diagnoses temperature through

$$T = \theta(p/p_{\mathrm{ref}})^{R_d/c_{pd}}.$$

The production algebraic pressure projection is the **dry rho–theta EOS**:

$$p = p_{\mathrm{ref}} \left(\frac{\rho_d R_d \theta}{p_{\mathrm{ref}}} \right)^{\gamma_d},$$
$$\gamma_d = \frac{c_{pd}}{c_{pd} - R_d}.$$

It is evaluated in double precision from the stored float density and potential temperature, then stored as float. This formula does not insert a moist gas constant or solve a new moist-mixture equilibrium. Moisture influences other parts of the model through phase sources, buoyancy/loading and diagnostic inventories. Replacing this EOS with a moist EOS would change the scientific model and its coupling contracts, not merely improve numerical precision. See the EOS implementation, its production application and the EOS tests.

The shared constants include $R_d = 287$, $R_v = 461.5$, $c_{pd} = 1004$, $c_{pv} = 1850$, $c_l = 4186$ and $c_i = 2106$, in J kg^{-1} K^{-1}; $T_r = 273.15$ K; $L_v = 2.5\times10^6$ and $L_f = 3.34\times10^5$ J kg^{-1}. These caloric constants are approximations rather than functions of temperature. Individual retained parameterizations may have their own pinned empirical constants. Changing those constants requires checking the process reference as well as the shared energy accounting. The constants header is the authoritative definition.

1.2 Resolved motion and the meaning of the grid

The dynamics are compressible and nonhydrostatic. Sound waves are retained, and vertical acceleration is not replaced by hydrostatic balance. A reference column is still useful: subtracting its pressure and density contributions lets the numerical vertical force resolve departures from the resting atmosphere without repeatedly balancing two large terms.

For orientation, the flat Cartesian vertical tendencies contain the following pressure, density, vapor-buoyancy and condensate-loading terms, evaluated at the appropriate staggered locations:

$$\begin{aligned} a_w = & -\frac{1}{\rho_d}\partial_z(p - p_0) - g\frac{\rho_d - \rho_{d0}}{\rho_d} \\ & + 0.608g(q_v - q_{v0}) - g(q_c + q_r + q_i + q_s + q_g + q_h). \end{aligned}$$

This is a description of the implemented approximation, not a derivation of a fully coupled moist Euler system. The vapor term is a virtual-temperature approximation; loading represents the weight of condensate. An additional potential-temperature buoyancy term must not be added on top of the existing perturbation-density force. Likewise, the mixture mass used in energy diagnostics does not authorize silently substituting mixture density in a pressure-gradient kernel that is defined using dry density. The Cartesian common helpers and C-grid dynamics make these choices explicit. Advection, diffusion, rotation and physical source terms complete the tendency; their discrete forms depend on the selected path.

On the C-grid, scalar quantities occupy cell centers and each velocity component occupies the face normal to that component. A face velocity therefore does not describe the same control volume as a cell-center scalar. Momentum transport, pressure work and kinetic energy must respect those different locations. Averaging face winds to the center and squaring that average loses kinetic energy associated with differences between adjacent faces. Nimbus's native face inventories use adjacent half-cell masses instead.

The collocated A-grid is a separate discretization. It has its own operators, boundary treatment and admitted configurations; it is not obtained by merely changing the interpretation of C-grid array indices. Storage also includes closure or halo locations. Only the physical cell domain belongs in a prognostic mass or energy sum. A finite number in a halo cell does not make that cell physical atmospheric volume.

For dry mass and transported scalar ϕ, the conservative target is

$$\partial_t \rho_d + \nabla\cdot(\rho_d u) = 0,$$
$$\partial_t(\rho_d \phi) + \nabla\cdot(\rho_d u \phi) = \rho_d S_\phi.$$

In finite-volume form, a scalar-cell update subtracts the signed sum of face fluxes from its inventory. The critical discrete requirement is that scalar transport use the density change and mass flux actually used by continuity. Otherwise a spatially uniform mixing ratio can change merely because its denominator and numerator were transported inconsistently. Nimbus retains the beginning density and interval mass flux for this recovery. Private particle fields declare whether they are stored per dry mass or per volume and are converted to the conserved representation before transport. See the mass-flux interface, Cartesian transport and private-field transport.

TVD reconstruction limits face states to suppress new oscillations. WENO5 combines candidate reconstructions according to local smoothness. Higher formal order does not remove the need for positivity, boundary and donor constraints, and it does not guarantee that a coarse grid resolves a storm feature. The permanent advection tests exercise the discrete operators rather than relying on visually plausible cloud output.

1.3 Terrain, frames and boundary conditions

Terrain-following coordinates change the physical volumes, gradients and face areas. They cannot be implemented consistently by changing only the plotted height of a Cartesian field. The mapped transport uses physical metrics and Jacobian-weighted inventories. Neighboring scalar centers, scalar layers and velocity dual volumes have distinct distances on a stretched or mapped grid.

An impermeable sloping ground has zero normal flow, not necessarily zero physical vertical velocity. For stationary terrain $z = h(x, y)$, the boundary condition implies $w = u\,h_x + v\,h_y$. Moving coordinates require the corresponding relative normal flux. This distinction also matters to kinetic energy: the diagnosed ground-face velocity has an owned half-cell contribution in energy basis v2. Terrain transport, mapped reconstruction tests and moving-energy tests cover parts of this contract. Broader moving/nested combinations still require their own admission and coupling evidence.

Boundary conditions are part of the physical problem. Periodicity identifies opposite faces; a closed boundary prevents normal transport; open and radiative boundaries approximate exchange with an unmodeled exterior. A Rayleigh or sponge layer relaxes state toward a selected reference and can do work on that state. It is not a transparent continuation of the interior equations. Surface

fluxes, open-boundary transport, sedimentation fallout and imposed relaxation must be included when interpreting domain budgets. An open, radiatively heated model should not be expected to conserve its internal domain energy without those exchanges. Boundary implementations and boundary tests specify the numerical realization.

Nested grids introduce another interface: parent and child must exchange inventories and fluxes, not just interpolate convenient field values. Changing a window's position also changes which physical atmosphere its arrays represent. Conservation, feedback, held sources and restart state therefore belong to the same movement transaction. See the detailed grid chapter.

1.4 Time integration and operator coupling

Atmospheric flow contains faster sound waves and slower advective and physical processes. The separation of acoustic modes is the central idea of Klemp and Wilhelmson (1978). Nimbus implements its own callback-based split sequence: slow tendencies are prepared, and pressure/density and momentum updates advance through acoustic substeps with boundary closure between the relevant stages. The configured acoustic-substep count is a minimum. The implementation increases it when the current sound-speed and physical-spacing bound requires more steps.

The broad idea is $y_t = F_{\text{slow}}(y) + F_{\text{acoustic}}(y)$, but that notation does not describe all finite-step errors. The order of pressure, velocity, boundaries, slow-stage reconstruction and physical sources affects the result. For that reason the split integrator and dynamics orchestration must be read together.

Separate admitted profiles use SSPRK3 or classical RK4. SSPRK3 takes forward Euler evaluations and convex combinations of stages; classical RK4 combines four tendencies with weights $1/6, 1/3, 1/3, 1/6$. Those order statements assume the right-hand side and intermediate state are treated consistently. Cadence-held physics, positivity constraints and discontinuous process switches can affect the order of the coupled model even when the standalone integrator passes an order test. Nimbus stages the coupled primary/private state and restores it after a failed stage; a partly accepted RK step is not a valid solution. Time-stepping tests and runtime timestep tests exercise the integration and stability policies.

The runtime considers advective, acoustic and applicable explicit-diffusion limits using physical grid spacings. A typical advective restriction depends on $\Delta t(|u|/\Delta x + |v|/\Delta y + |w|/\Delta z)$; the actual policy includes the selected operator and mapped geometry. Vertical implicit diffusion uses a backward-Euler tridiagonal solve for a density-weighted flux-divergence equation. Its stability does not justify an arbitrarily large timestep for accuracy or for the other coupled processes. Radiation and boundary-layer refresh cadences are additional time scales: the runtime integrates their held source over the appropriate interval.

1.5 Energy: an inventory is not automatically an invariant

Nimbus retains a legacy energy proxy for compatibility and a versioned moist-mixture inventory. The latter is the relevant basis for current energy attribution. Per kg of dry air, its thermodynamic part is

$$\begin{aligned} u = {} & c_{vd}(T - T_r) \\ & + (c_{vv} q_v + c_l q_l + c_i q_f)(T - T_r) \\ & + (L_v - R_v T_r) q_v - L_f q_f, \end{aligned}$$

where $q_l = q_c + q_r$, $q_f = q_i + q_s + q_g + q_h$, $c_{vd} = c_{pd} - R_d$ and $c_{vv} = c_{pv} - R_v$. The liquid reference is at freezing; frozen material has the negative fusion reference, and the vapor reference converts the chosen latent enthalpy to internal energy. Condensed-phase pressure-volume work is neglected. A local helper may omit the constant dry reference when dry mass is fixed; that leaves its source increment unchanged, but reference offsets must still be treated consistently when mass crosses a domain boundary.

The complete diagnostic adds resolved kinetic and gravitational energy using mixture mass, and subgrid TKE using dry mass. Basis v2 preserves these choices and includes the diagnosed mapped ground-face kinetic contribution. The inventory contract and implementation are more precise than a single plotted "total energy" curve.

Computing this inventory does not make every dynamics/physics operator conserve it. In particular, pressure is still projected with the dry EOS. Applying that projection can change diagnosed temperature and energy. The model reports the change instead of assuming it vanishes. A finite velocity increment also has a quadratic kinetic contribution:

$$\Delta K = mu \cdot \Delta u + \tfrac{1}{2} m |\Delta u|^2$$

when the associated mass is fixed. Requested linear work, the finite-step correction, and float-storage effects are distinct quantities. If mass changes, its contribution must also be included. The energy-budget tests check these distinctions, including physical domains and mapped geometry.

A closed accounting residual means that attributed changes reproduce the committed state change. It is not by itself proof that the physical energy transfer is correct. Physical validation additionally requires the chosen source law, boundary work, transport, phase references and thermodynamic coupling to agree. The incomplete source/EOS work described below preserves this distinction explicitly.

1.6 Bulk clouds and precipitation

Bulk microphysics follows category masses and selected moments of particle-size distributions instead of individual droplets and crystals. Kessler is a warm-rain family; Lin, Thompson and Milbrandt contain different mixed-phase categories and closure assumptions. Selecting one changes the physical model, not just numerical cost. Fall-speed laws, conversion efficiencies, freezing, activation and collection are scheme-specific. Nimbus's Thompson implementation also requires pinned activation, collection and cold-process tables; loading a different table is a scientific input change.

For the internal gamma-distribution moment calculations, write a spectrum per kg dry air as $n(D) = N_0 D^\alpha e^{-\lambda D}$. Its moments satisfy

$$\begin{aligned} M_k &= N_0 \frac{\Gamma(\alpha + k + 1)}{\lambda^{\alpha+k+1}}, \\ N &= M_0, \qquad q = c M_d, \qquad Z = M_6 \end{aligned}$$

for a particle mass law $m(D) = cD^d$. The assumed spectrum makes a finite set of prognostic variables sufficient to evaluate many integrals. It also means that not every positive q/N/Z triplet represents a realizable spectrum. The third moment is a diameter-sixth moment in this convention; it must not be confused with every plotted radar product or with a per-volume concentration. Storage-unit adapters and the radar forward operator supply those conversions.

Milbrandt and Yau (2005, Part II) propose the additional moment closure and distinguish collection, initiation and category-conversion contributions. Nimbus's retained internal source closure uses, for appropriate processes,

$$\frac{\dot{Z}}{Z} = \frac{6}{d}\frac{\dot{q}}{q} + \left(1 - \frac{6}{d}\right)\frac{\dot{N}}{N}.$$

This is a process-rate relation under its spectral assumption, not permission to apply an arbitrarily large finite increment. Original donor bounds and endpoint realizability are checked separately. Snow's retained mass law is non-cubic, so replacing every exponent by three would change the calculation. The moment inversion, sedimentation and independent moment tests provide the concrete conventions.

Sedimentation transports particles vertically relative to the air. Its mass, number and higher moments can have different weighted fall speeds. The conservative inventory uses dry density and actual layer thickness, with surface loss booked as fallout. Source/sedimentation subcycling controls bounded intervals; it does not make a terminal-velocity approximation a resolved particle-dynamics calculation. Phase changes redistribute water and change heat through the selected closure. A scalar nonnegativity clamp alone cannot establish water, heat and particle-moment consistency.

1.6.1 Guarded triple-moment surface work

Public Milbrandt triple-moment admission remains **Engineering incomplete**. The internal source/EOS and wet-hail helpers are bounded research implementations, not public replacements for ordinary double-moment behavior. Their assumptions must remain visible when the historical experiments are archived.

The wet-surface helper chooses a zero-storage interface at 0°C, with no prognostic hail-core temperature or retained liquid film. Incoming collected liquid/solid and converted original hail begin at the cell temperature; surface outputs subsequently return to the represented cell temperature. With signed vapor C, conduction H, collected liquid L, collected frozen mass F, freezing R and original-hail melting M, it uses

$$H + L_v C + c_l(T - T_r)L + c_i(T - T_r)F + L_f R = [L_f - c_i(T - T_r)]M.$$

It requires positive remaining liquid, keeps original donors separate from newly created material and rejects unowned transitions. Explicit vapor mass and this material-temperature path extend the Lin et al. (1983) wet-growth approximation, whose equation-43 discussion neglects vapor mass. They must not be presented as a literal transcription of that paper.

An opt-in internal composition shares original vapor with zero-riming snow deposition, then solves cloud saturation at the subsequent EOS energy endpoint. Its surface rates use the entry state; new equilibrium cloud cannot fund those captured collection rates. The cell energy mapping preserves the no-source EOS reference and reports projection and storage work. Its material-enthalpy comparison subtracts the gas pressure-volume change $\Delta[(R_d + R_v q_v)T]$ for the held dry-mass cell.

This is a declared model-currency conversion, not a completed global mechanical/thermal transport closure.

The current full joint fixtures cover wet freezing and both signs of surface and cloud exchange. Saturating shared-vapor limits are tested at helper level; accepted full cap-active intervals, new joint melting/time-convergence, remaining vapor/collection families and the live field/source→EOS transaction remain unfinished. See the source/EOS owner, shared-vapor owner and permanent joint tests.

1.7 Unresolved mixing, surface exchange and radiation

Resolving a grid cell does not resolve all turbulent motion inside it. The LES distinction between resolved large eddies and parameterized smaller motions is central to Moeng (1984). Nimbus's standard Smagorinsky branch forms an eddy viscosity proportional to $(C_s\Delta)^2|S|$, where S is the resolved strain tensor. Its dynamic and stability-dependent variants add their own assumptions. A grid-derived filter width represents the unresolved scale; a configured physical filter width changes that model even when the grid is unchanged.

The prognostic TKE closure uses an equation of the form $e_t = P_s + P_b - \epsilon + \text{transport}$. Its retained viscosity is $0.1\ell\sqrt{e}$; stable stratification can limit ℓ relative to the grid filter width. Production, dissipation and scalar diffusivities are a coupled closure, not independent tuning knobs. Tensor locations, dry-density weights and boundary work matter to the discrete energy exchange. The TKE implementation and SGS tests define Nimbus's choices.

Boundary-layer schemes represent vertical column mixing and surface exchange at a different level of approximation. Slab, YSU and MYNN are distinct models. YSU includes nonlocal transport and entrainment; MYNN uses TKE-dependent stability relations. These schemes publish realized column sources over their refresh interval. Combining PBL and resolved-grid SGS mixing requires an explicit compatible profile so that unresolved transport is not simply counted twice. Surface-flux and column assumptions belong to the chosen closure, not to the geometric boundary-condition dispatcher. Column evaluation and boundary-layer tests are the relevant source and checks.

Radiation changes atmospheric energy through flux divergence. For a column with upward-positive net radiative flux, the local heating relation has the form $\dot{T} = -(\rho_d c_p)^{-1}\partial_z F_{\text{net}}$, followed by the runtime's temperature-to-potential-temperature conversion. The simple-grey models make broadband idealizations; the absorbing water/cloud-path option assumes pure absorption and does not model cloud scattering. Native RRTMG uses explicit AER coefficient data, reduced spectral quadrature and separate longwave/shortwave transfer. Nimbus represents resolved condensate as fully cloudy within the relevant cell; that is not a subgrid cloud-fraction or overlap model.

Solar geometry, optical data, surface temperature/albedo and update cadence are scientific inputs. A terrain shadow or horizontally transporting shortwave operator introduces additional geometric and angular assumptions beyond an independent vertical column. Its availability cannot be inferred from the existence of the column solver. Consult radiation runtime, RRTMG optics and radiation tests with the actual catalog scope.

1.8 Precision, reproducibility and scientific interpretation

The main fields are stored as float, while many source integrals, EOS calculations, inventories and reductions use double precision. A double calculation followed by float storage can still change a

budget. Those storage changes are observable contributions, not automatically physical sources. The deterministic budget path reduces fixed work partitions in a fixed order; ordinary unordered parallel reductions need not be bitwise identical when the worker count changes. Hardware/backend parity and deterministic reduction are separate contracts.

Tests of a formula, conservation identity, rejected input, restart and native backend establish different things. A checked reference column does not prove storm-scale accuracy. An exact restart does not prove an untested process combination. A finite residual does not excuse an unexplained source, and a zero residual does not prove a correct physical transfer. Changes should be judged against the declared units, assumptions, boundary work and independent reference appropriate to the affected operator.

The maintained status vocabulary is **Engineering incomplete**, **Ready**, **Campaign validated**, **Externally reviewed** and **Fully qualified**. Readiness requires complete engineering, bounded tests, public selection and ordinary admission for the stated scope. Long campaigns and independent scientific review are additional requirements for full qualification. The repository contract defines these distinctions. Numerical results should identify the configuration, coefficient inputs, precision and exact supported profile rather than relying on the name of a parameterization.

Chapter 2

Initial conditions, soundings and prescribed environments

An initial condition specifies the experiment from which the equations evolve. It includes atmospheric thermodynamics, wind, geometry and any deliberate trigger. Those choices determine the initial inventories and available energy; they are not neutral preparation for a later physics calculation. Nimbus separates a sounding source, a hodograph source and a trigger so that choosing microphysics does not secretly choose an unrelated environment or disturbance.

2.1 Three independent questions precede the first timestep

The sounding source supplies a column of pressure, temperature/theta, moisture and possibly wind. A file reader interprets a supplied profile; a parametric source constructs an idealized one. The hodograph policy decides whether to retain available file winds or build a separate environmental wind. Finally, the trigger applies no disturbance, a warm bubble or a vortex seed. These are separate mathematical decisions even when a preset chooses all three.

The sounding contract, hodograph contract and trigger contract provide the interfaces. Profile selection and the configuration compiler connect them to the ordinary runtime. A helper's existence does not imply that every option it accepts is available through an admitted public profile.

An initial environment also needs physical coordinate locations. A terrain column may begin above the datum of an observed sounding, and its scalar and velocity levels need not coincide. The initialized pressure, density and theta must be interpreted through the production dry rho–theta relation. Moist parcel diagnostics may use virtual temperature while the dynamical EOS uses a dry pressure law; these are distinct approximations whose roles must be stated.

2.2 Reading a sounding is a unit and representation conversion

The public reader catalog offers `auto`, `cm1` and `sharpy`. Automatic selection examines content and file signatures. The reader family includes text and supported netCDF/HDF5 profile forms; a filename extension alone does not identify a valid physical sounding. The sounding module validates the supplied variables, finite values, ordering, units, dimensions and model coverage before a profile is used.

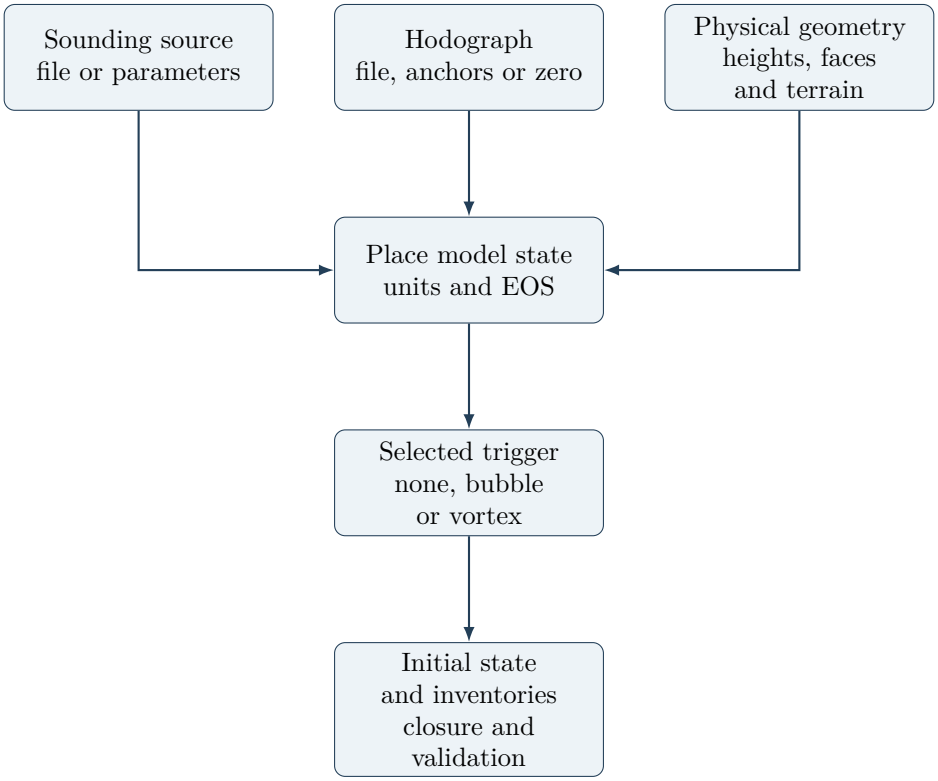


Figure 2.1: Logical responsibilities in constructing an experiment. Environmental thermodynamics, wind and geometry are separate inputs. A prescribed trigger can change the initial water and energy inventories; it is not automatically a conservative microphysical transfer. Exact execution and admission belong to the initialization and configuration owners.

The mathematical input is a set of values at stated heights or pressure levels, not a picture of a skew-T diagram. For example, a value written in grams of vapor per kilogram of dry air must be divided by 1000 to become the solver's kg/kg dry-air mixing ratio. A temperature in degrees Celsius must be converted to kelvin before evaluating an exponential saturation formula. Confusing a moist-air specific humidity $q_s = m_v/(m_d + m_v)$ with a dry-air mixing ratio $r_v = m_v/m_d$ gives the wrong quantity even though both are dimensionless:

$$r_v = \frac{q_s}{1 - q_s}, \qquad q_s = \frac{r_v}{1 + r_v}.$$

A reader must know which definition the source supplies rather than infer it from a small decimal value.

Strict-v1 file initialization retains automatic format routing, linear interpolation, no extrapolation and no silent procedural fallback. The lower-level fallback source remains compatibility code: its existence does not authorize replacing a missing observation with an invented atmosphere under the ordinary strict path. Sounding coverage of the required model heights is therefore a real input requirement.

For a scalar quantity a between valid source levels z_j and z_{j+1}, linear interpolation uses

$$a(z) = (1 - \lambda)a_j + \lambda a_{j+1}, \qquad \lambda = \frac{z - z_j}{z_{j+1} - z_j}.$$

This preserves the endpoint values and is bounded between them for $0 \le \lambda \le 1$. It is not a conservative remapping of a three-dimensional mass inventory. That is acceptable only because it is being used to define a new initial profile, with its resulting inventories diagnosed afterward. Interpolating Cartesian wind components directly also avoids interpolating across the discontinuity between directions near 0 and 360 degrees.

2.3 Hydrostatic preparation and the model EOS must be distinguished

For an idealized resting column, hydrostatic balance is $dp/dz = -\rho g$. Combining an ideal-gas relation with a chosen temperature profile gives an integral pressure relation, schematically

$$\log \frac{p(z_2)}{p(z_1)} = -\int_{z_1}^{z_2} \frac{g}{R_d T_v(z)}\, dz$$

when the chosen preparation uses virtual temperature T_v. The dry version uses T instead. This equation states the assumptions used to build a profile; it does not replace the solver's declared dry-density pressure law or prove that a sampled profile is a discrete resting state.

Nimbus's parametric source constructs pressure along its idealized column and diagnoses thermodynamic quantities from that construction. File input has its own conversion and validation. Placement onto model fields, reference-state construction and later EOS projection must remain consistent with their respective definitions. A continuous hydrostatic relation and the discrete pressure-gradient/gravity pairing are separate checks, as developed in the grid chapter. A small initialization adjustment wave can reflect a mismatch between them rather than an instability of the supplied atmosphere.

2.4 CAPE targeting is a bounded inverse problem

The `parametric_cape` source builds an idealized mixed layer, a free-tropospheric temperature lapse and a stable upper cap. Its moisture profile uses the configured near-surface amount, a vertical decay and a relative-humidity limit. The control variable used to fit CAPE is the free-tropospheric lapse rate, not an arbitrary independent adjustment of every level.

A surface-parcel buoyancy diagnostic has the conceptual form

$$B(z) = g\frac{T_{v,p}(z) - T_{v,e}(z)}{T_{v,e}(z)}, \qquad \mathrm{CAPE} = \int_{\mathrm{LFC}}^{\mathrm{EL}} \max(B, 0)\, dz.$$

The parcel and environment must use the same pressure and humidity conventions. Nimbus's diagnostic estimates the lifting-condensation level, lifts the parcel dry below it and with a retained moist lapse approximation above it, and integrates resolved positive buoyancy. It also reports inhibition and the associated characteristic heights. These quantities depend on the diagnostic algorithm and vertical resolution, not just the name of the sounding.

The source bisects a lapse-rate interval from 0.0055 to 0.0098 $\mathrm{K\,m^{-1}}$, seeking agreement within 50 $\mathrm{J\,kg^{-1}}$, with at most 60 iterations. These are the current implementation's search bounds and stopping rule. The final profile is still returned after the iteration limit; the source alone is not a proof that every requested target is realizable within that family. Requested and realized CAPE must be distinguished and the applicable diagnostics/admission checked. See the CAPE source and parcel diagnostics.

The `parametric_targets` branch translates a requested diagnostic set into parameters for that same family. Surface moisture may be inferred from an LCL target; a mixed-layer cap provides an inhibition control; requested LFC/EL values influence the mixed-layer and upper-profile construction. Those controls are coupled. CAPE, CIN, LCL, LFC and EL cannot generally be prescribed as five independent exact numbers by this limited profile family. The target translator contains heuristics and explicit input bounds; the resulting diagnostics remain the report of what was actually constructed.

Weisman and Klemp (1982) supplies important context for studying buoyancy and shear in idealized convective environments. A preset inspired by that literature is not a claim that Nimbus reproduces the paper's entire environmental construction, numerical model or storm evolution.

2.5 Wind profiles and hodograph precedence

A hodograph is the path traced by $(u(z), v(z))$ with height. Its differences control vertical shear; its curvature affects storm-relative flow and rotation diagnostics. Translating every wind vector by a constant changes storm-relative interpretation unless the storm-motion reference is translated consistently. It does not change the environmental vector shear between two levels.

Nimbus's `auto` hodograph policy uses sounding winds when they are available. Otherwise it constructs the configured parametric profile. `wk_param` explicitly selects the parameterized wind, and `zero` supplies a calm column. A file containing valid winds and an explicit separate hodograph are therefore different experiments from the automatic file-wind path.

The retained `wk_param` implementation is piecewise linear in each Cartesian component through anchors at the surface, 1 km and 6 km. Above 6 km it holds the final wind constant. Between

anchors a, b it uses

$$(u, v)(z) = (1 - \lambda)(u, v)_a + \lambda(u, v)_b, \qquad \lambda = \frac{z - z_a}{z_b - z_a}.$$

The selector name is not a promise of an arbitrary smooth quarter-circle or a complete reproduction of a published hodograph. The implemented interpolation and wind precedence are explicit. Sampling the environmental wind at the chosen scalar/face positions belongs to initialization; it must not silently change the staggering contract used by later transport.

2.6 The warm bubble changes both heat and, conditionally, water

The current Cartesian bubble helper uses distance r from its specified physical center and a compactly applied Gaussian factor

$$f(r) = \begin{cases} \exp(-9r^2/R^2), & r \le R, \\ 0, & r > R, \end{cases} \qquad \Delta\theta = \Delta\theta_{\max} f(r).$$

Thus $R/3$ is the Gaussian width in this convention, and the field is cut off at R rather than extending to infinity. The helper retains a minimum radius of 100 m and uses physical terrain height where active metrics are present. Public validation and exact-profile admission additionally govern which settings can reach that helper.

When pressure and vapor fields have matching shape and the applied theta increment is positive, the helper scales vapor to retain its local saturation ratio at the held pressure:

$$q_v^{\text{new}} = q_v^{\text{old}} \frac{q_{vs}(T^{\text{new}}, p)}{q_{vs}(T^{\text{old}}, p)}.$$

The ratio is applied only when the old saturation amount exceeds the retained small-value threshold. The helper uses its local liquid-water saturation fit and an Exner exponent of 0.2857 for this conversion. Those are its actual approximations; one should not assume that it calls the selected microphysics scheme's saturation routine or performs a closed moist adjustment.

Optional cloud seeding separately adds a configured cloud-water amount times $f(r)$. It is not subtracted from vapor as a condensation transaction. The warm bubble therefore changes the initial thermal state and can add vapor and cloud water. These are prescribed initial-condition changes. Their resulting mass and energy belong in the initial inventory, rather than being mislabeled as a conservative source during the first timestep. They also do not guarantee that every chosen environment will produce sustained convection.

See the Cartesian initializer. A trigger named `none` makes no such disturbance; choosing it does not remove environmental instability or subsequent stochastic forcing selected elsewhere.

2.7 The Rankine seed has a defined profile and an implementation gap

The `vortex_seed` helper adds an azimuthal velocity around a horizontal center. With radius r_m of maximum wind and signed peak speed V_m,

$$v_\vartheta(r) = \begin{cases} V_m r / r_m, & 0 < r \le r_m, \\ V_m r_m / r, & r > r_m, \end{cases} \qquad \Delta u = -v_\vartheta \frac{y - y_c}{r}, \quad \Delta v = v_\vartheta \frac{x - x_c}{r}.$$

The axis receives no singular increment. The source adds this wind to the existing environmental field and applies a hard upper-height cutoff at `z_top_m`. It does not construct an accompanying balanced pressure perturbation. Sampling a continuous rotational formula also does not, by itself, prove zero discrete divergence or exact balance on the selected staggered grid.

The public configuration currently parses and serializes `center_z_m`, but this helper does not use it to place the vortex vertically. Its applied placement uses the horizontal center and the upper cutoff described above. This is a known initialization implementation gap; the parameter must not be interpreted as an implemented vertical translation. Exact admitted profiles still have their own restrictions, so this description is not blanket admission of every vortex configuration. The concrete owner is the vortex-seed implementation.

2.8 Terrain shapes are also initial scientific inputs

The terrain catalog provides `none`, `bell` and `schar`. The flat choice does not imply uniformly spaced vertical levels; the geometry owner can still provide stretched physical coordinates. A two-dimensional bell ridge uses

$$h(x) = \frac{h_0}{1 + (x/a)^2},$$

while the axisymmetric bell uses $h(x, y) = h_0 a^3/(a^2 + x^2 + y^2)^{3/2}$. The Schär profile is

$$h(x) = H \exp[-(x/a)^2] \cos^2(\pi x/\ell).$$

Height and width parameters have physical length units, and the source supplies the corresponding slopes. These shapes define geometric input, not a physical land-surface evolution model.

The mapping must turn that topography and the vertical grid into positive physical layer volumes and compatible face metrics. A wind tangent to a slope, an environmental profile sampled at the actual height, a surface flux per true area and a radiation shadow are all consequences of the same terrain geometry. They must not each reconstruct a different terrain from nominal spacing. See the shape functions and the detailed grid chapter.

2.9 What should be recorded about an initialized experiment

Retain the source profile or parameter values, wind-selection precedence, physical grid and terrain, trigger settings, stochastic seed/member identity, requested diagnostics and realized diagnostics. Record the initial dry mass, water and energy basis after all prescribed interventions and closures. The same file profile interpreted on a different grid, with a different trigger or a different moisture convention is a different experiment even when its name is unchanged.

The initial-condition and sounding tests under tests/init exercise parsing, thermodynamic conversion, coverage, selection and trigger independence. The first evolution step then checks a different contract: it must advance that accepted state through the selected numerical and physical owners. Restart is different again; it restores a continuation state and should not rerun an initial bubble or invent a new sounding in place of saved scheme-private history.

Chapter 3

Grid, resolved dynamics and the coupled timestep

This chapter develops the grid and time-integration arguments behind the mathematics overview. It describes the Cartesian atmosphere implemented in Nimbus. Equations written for a uniform, stationary grid are introduced as explanatory cases; the mapped operators must additionally use their physical metrics. The implementation supports several discrete paths, and an equation in this chapter does not imply that every configuration or backend combination is admitted.

3.1 Start with an inventory, not an array value

A water mixing ratio is an amount of water divided by dry-air mass. It is not itself the conserved water amount in a cell. If cell c has physical volume V_c, its dry mass and the inventory of a scalar ϕ are

$$M_c = \rho_{d,c} V_c,$$
$$Q_c = M_c \phi_c.$$

For water, ϕ is a species mixing ratio. For a particle number stored per kg dry air, Q_c is the particle count. A number stored per cubic metre would instead have inventory $V_c N_c$. These distinctions explain why the private microphysics transport interface declares units: applying an identical array update to different units would solve different conservation equations.

Integrate dry-mass continuity over the cell and apply the divergence theorem. Let F_f denote outward dry-mass flow through face f, in kg s^{-1}, averaged over the interval. A stationary finite-volume update is

$$M_c^{n+1} = M_c^n - \Delta t \sum_{f \in \partial c} F_f,$$
$$Q_c^{n+1} = M_c^n \phi_c^n - \Delta t \sum_{f \in \partial c} F_f \phi_f + \Delta Q_{c,\text{source}},$$
$$\phi_c^{n+1} = Q_c^{n+1} / M_c^{n+1}.$$

An internal face has one flow and opposite signs in its two neighboring cells. Its contribution

therefore cancels from the domain sum. At a physical boundary it remains an exchange with the exterior. This cancellation, and the use of the same flow in both equations, are the useful conservation properties.

For example, set every ϕ to a constant ϕ_0 and omit sources. Consistent reconstruction returns $\phi_f = \phi_0$, so the second equation becomes $Q_c^{n+1} = \phi_0 M_c^{n+1}$. A uniform mixing ratio stays uniform even while density changes. If scalar transport recomputes a different mass flow, that identity no longer holds. Nimbus therefore retains beginning density and the continuity interval flux used to recover transported scalars. The relevant contracts are mass flux, Cartesian advection and private scalar units.

Array bounds are not physical integration bounds. Arrays include storage used for closure and halos, and boundary values can be finite and quite plausible. Including them in a mass sum would nevertheless count extra atmosphere. The physical cell domain defines the half-open scalar ranges that belong to the atmosphere. Operators must then identify the faces bounding those cells, rather than infer the physical domain from a loop over every allocated element.

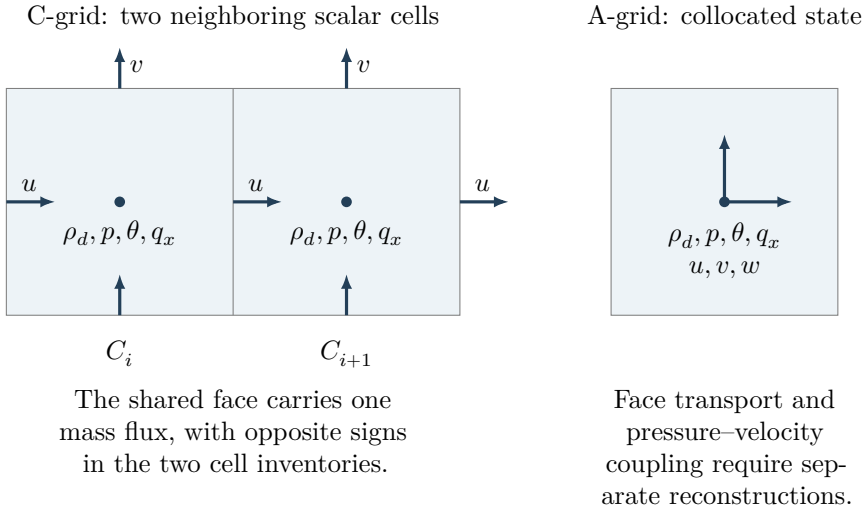


Figure 3.1: Horizontal plan views of the two layouts. On the C-grid, w is stored on the horizontal faces above and below each scalar volume and is omitted from this plan view. The labels describe physical locations, not a universal array index convention. Halo and closure locations are not additional atmospheric cells.

3.2 Why the locations of pressure and velocity matter

On the Cartesian C-grid, p, ρ_d, θ and water quantities are scalar cell values. Velocity components lie on their respective normal faces. In the array convention, $u(i, j, k)$ is the east face of scalar cell (i, j, k), $v(i, j, k)$ is its north face, and $w(i, j, k)$ is its upper face. The lower ground face has a separately owned boundary value. An index alone therefore does not identify one common physical point for every field.

For two adjacent uniform-grid cells, the x-face dry-mass flow is constructed from a density average and the already face-located wind:

$$F_{i+1/2} = A_{i+1/2} \frac{\rho_{d,i} + \rho_{d,i+1}}{2} u_{i+1/2}.$$

Here A is physical face area. The mass-flux helper returns the product before area integration. The pressure difference $(p_{i+1} - p_i)/\Delta x$ is located at this same face, so the pressure acceleration and normal velocity naturally share a location. Density interpolation and the perturbation-pressure choice remain part of the momentum operator; placement alone does not prove an energy identity.

The A-grid stores velocity at scalar locations. Its uncorrected face product uses both interpolated density and interpolated velocity:

$$F^{\text{raw}}_{i+1/2} = A_{i+1/2} \frac{\rho_{d,i} + \rho_{d,i+1}}{2} \frac{u_i + u_{i+1}}{2}.$$

That interpolation introduces a familiar pressure problem. Consider the alternating pattern $p_i = p_* + a(-1)^i$. Its centered cell gradient is zero:

$$\frac{p_{i+1} - p_{i-1}}{2\Delta x} = 0, \qquad \frac{p_{i+1} - p_i}{\Delta x} \neq 0 \quad (a \neq 0).$$

Thus a collocated cell-gradient operator can fail to see an oscillatory pressure field that should drive face motion. Nimbus's `CollocatedPressureCoupling` corrects the face velocity using the direct pressure jump minus the interpolated cell pressure acceleration. Continuity and retained scalar transport receive the same correction. This coupling is deliberately inactive on the C-grid, whose face velocity is already paired with an adjacent-cell pressure jump. The two grids consequently have distinct numerical operators, not merely different names for the same arrays.

Face placement also changes kinetic-energy accounting. Let M be a scalar cell's moist-mixture mass and consider only its two x faces. Assigning each face half the cell mass gives

$$\begin{aligned} K_{x,\text{faces}} &= \frac{M}{4}(u_L^2 + u_R^2), \\ K_{x,\text{center}} &= \frac{M}{2}\left(\frac{u_L + u_R}{2}\right)^2, \\ K_{x,\text{faces}} - K_{x,\text{center}} &= \frac{M}{8}(u_R - u_L)^2. \end{aligned}$$

Averaging first and then squaring loses the positive contribution from within-cell face variation. Summing the analogous x, y and z terms is the scalar-cell representation of Nimbus's C-grid kinetic inventory. An interior face gathers half-cell mass from each side; a physical boundary contributes only its owned side. This is why an A-grid kinetic-energy formula cannot be substituted into a C-grid diagnostic.

The mixture mass in this diagnostic is $M = \rho_d V(1 + q_t)$. It does not change the density convention of the dynamical pressure acceleration. The latter uses its declared dry-density approximation. The momentum-flux owner and energy-budget implementation must be assessed together when asking whether a particular discrete force and transport preserve a chosen energy inventory.

3.3 What reconstruction adds to the flux equation

The finite-volume equations require a face scalar ϕ_f. A cell average does not uniquely determine that value. First-order upwinding uses the donor cell value; it is robust under an appropriate timestep restriction but spreads sharp features. Higher-order reconstruction estimates subcell variation. TVD limiting reduces a reconstructed slope when neighboring data indicate an extremum or discontinuity. WENO5 combines candidate reconstructions with weights that respond to their local smoothness, reducing the influence of a stencil that crosses a sharp feature.

These are assumptions about what can be inferred between stored samples. They do not add resolved information below the grid spacing. Their formal order applies in sufficiently smooth regions, with compatible boundaries and time integration. Limiters, positivity enforcement and process thresholds can reduce local order. A narrow precipitation shaft can therefore be sensitive to resolution even when its integrated mass is well conserved.

Momentum has its own control volumes and transport paths. The conservative C-grid momentum owner builds dual-volume fluxes from dry-density transport. Other retained slow-tendency paths use derivative-based advection. It would be incorrect to describe every selectable path as one identical finite-volume discretization of the complete moist Euler equations. The current configuration and advection tests identify the actual operator being exercised.

3.4 Pressure, gravity and the fast acoustic response

Potential temperature provides a convenient adiabatic thermal variable. With the retained constant dry-air parameters, Nimbus uses

$$
\begin{aligned}
T &= \theta \left(\frac{p}{p_{\mathrm{ref}}} \right)^{\kappa_d}, & \kappa_d &= R_d / c_{pd}, \\
p &= p_{\mathrm{ref}} \left(\frac{\rho_d R_d \theta}{p_{\mathrm{ref}}} \right)^{\gamma_d}, & \gamma_d &= \frac{c_{pd}}{c_{pd} - R_d}.
\end{aligned}
$$

These relations imply $p = \rho_d R_d T$. Moisture is not inserted into this pressure relation as an additional gas constant. That is a deliberate retained model convention. Phase heating, vapor buoyancy, condensate loading and the moist diagnostic energy are separate parts of the implementation.

To see why this system supports fast waves, linearize about a uniform resting dry state, holding θ fixed during the acoustic perturbation. Then $p' = c_s^2 \rho_d'$ with $c_s^2 = \gamma_d p_0 / \rho_{d0}$. Linear continuity and momentum give

$$
\begin{aligned}
\partial_t \rho_d' + \rho_{d0} \nabla \cdot u' &= 0, \\
\partial_t u' &= -\rho_{d0}^{-1} \nabla p', \\
\partial_{tt} \rho_d' &= c_s^2 \nabla^2 \rho_d'.
\end{aligned}
$$

The final equation is the acoustic wave equation for this simplified limit. It explains why an advectively acceptable large step may still be unstable for pressure waves. Nimbus's acoustic policy increases the configured minimum number of substeps using current sound speed and minimum physical spacing. The retained estimate uses a target acoustic Courant number of 0.5 and an additional substep beyond the ceiling estimate. This is an implementation stability policy, not proof of accuracy for an arbitrary physical timestep.

In a stratified atmosphere, pressure and gravity are each large. A reference column satisfies the intended background balance, and perturbation operators resolve the departure from it. For the flat C-grid vertical force, the retained terms have the form

$$\begin{aligned} a_w = & -\frac{1}{\rho_d}\partial_z(p-p_0) - g\frac{\rho_d-\rho_{d0}}{\rho_d} \\ & + 0.608g(q_v - q_{v0}) - g(q_c + q_r + q_i + q_s + q_g + q_h), \end{aligned}$$

with quantities interpolated to their velocity location. The perturbation pressure force belongs to the fast update; the other displayed contributions belong to the slow preparation. The vapor term approximates virtual-temperature buoyancy, while condensate contributes loading. Adding a separate thermal buoyancy formula on top of the existing perturbation-density term would change the model and can double-count the thermal effect. See the C-grid tendency implementation and shared Cartesian force helpers.

3.5 Terrain and moving coordinates change the geometry

Let a terrain-following mapping be $z = z(x, y, \zeta)$. Even though arrays remain rectangular in index space, physical cells need not have equal volumes or orthogonal faces. The coordinate Jacobian changes the amount of atmosphere represented by a coordinate-space volume. Physical gradients require metric terms, and the normal velocity through a mapped face is generally a combination of Cartesian velocity components.

For a stationary lower boundary $z = h(x, y)$, impermeability says that the wind is tangent to the surface. Taking the material derivative of $z - h(x, y) = 0$ gives

$$w = u\,\partial_x h + v\,\partial_y h.$$

Thus zero normal flow can coexist with nonzero physical vertical velocity. Energy basis v2 includes the diagnosed ground-face kinetic contribution, rather than treating this face as implicitly motionless. On stretched layers, scalar center distances, layer thicknesses and velocity dual-volume widths must also be distinguished. A single nominal Δz is not an adequate substitute.

A moving control volume changes the transport velocity. Reynolds transport gives the relevant normal flux as

$$F_f = \rho_{d,f}(u_f - v_{g,f})\cdot n_f\, A_f,$$

where v_g is grid velocity. Nimbus retains Earth-frame wind and subtracts grid translation when constructing the relative flux. Adding grid velocity to every stored wind during a window shift would change the physical state. Earth-fixed terrain and terrain attached to a translating computational frame also pose different boundary problems.

Window movement and nesting require conservative remapping of inventories, consistent velocity locations, and ownership of boundary/feedback exchanges. Interpolating only visible water and wind fields would leave private particle moments, held physical sources and restart state inconsistent. The terrain transport interface, mapped reconstruction tests, moving-energy tests and the detailed grid chapter describe these additional contracts.

3.6 How the split timestep is assembled

Klemp and Wilhelmson (1978) provide the physical and numerical motivation for separating compressible acoustic motion from slower cloud-scale evolution. Nimbus implements a specific callback sequence rather than reproducing an entire published model unchanged. The following describes its split-explicit path; SSPRK3 and RK4 have separate stage owners.

At the outer runtime level, radiation and boundary-layer sources are refreshed when their cadence requires it. Held sources remain physical state with an interval of validity. The dynamics owner prepares slow tendencies, including applicable advection, forcing and buoyancy. Where the configured slow-stage method requires a predictor, it constructs the corresponding tendency average and restores the authoritative stage state before the final advance.

The split integrator first applies the slow increment over the large timestep. It seeds radiative acoustic boundary history from that post-slow state. Each acoustic substep then advances pressure/density, closes the pressure/density boundaries needed by the next operation, and advances momentum using the newly available pressure and density. Boundary time advancement has its own owner; a final zero-duration closure must not advance a radiative boundary a second time. Native batched implementations must preserve this sequencing even when their implementation does not make one host call per substep.

The production CPU pressure commit updates dry density through continuity, stores the new float density and diagnoses pressure through the dry rho–theta EOS from the stored density and potential temperature. A pressure tendency buffer alone is therefore not the complete description of the committed pressure state. This distinction matters when evaluating pressure work and storage-rounding contributions.

After the acoustic sequence, the runtime recovers thermodynamic and tracer transport using the retained continuity flux when the selected path has not already coupled that transport. It also transports declared private microphysics state. Microphysics then produces and commits its source changes. Applicable explicit SGS and diffusion follow, and a final pressure projection restores the algebraic rho–theta EOS relation. Sponge and final boundary operations have their subsequent ownership. The exact conditionals belong to dynamics orchestration, the split integrator and headless runtime.

This order has a scientific consequence: a private microphysics cell helper that assumes its source is followed immediately by EOS projection cannot be inserted into the live solver without accounting for intervening SGS/diffusion and other state changes. The guarded source/EOS transaction is tested as an explicit endpoint mapping; its complete field-level runtime integration remains Engineering incomplete.

For comparison, an autonomous SSPRK3 system $y_t = L(y)$ has the standard convex stage structure

$$\begin{aligned} y^{(1)} &= y^n + \Delta t L(y^n), \\ y^{(2)} &= \tfrac{3}{4} y^n + \tfrac{1}{4}[y^{(1)} + \Delta t L(y^{(1)})], \\ y^{n+1} &= \tfrac{1}{3} y^n + \tfrac{2}{3}[y^{(2)} + \Delta t L(y^{(2)})]. \end{aligned}$$

The corresponding final tendency weights are 1/6, 1/6 and 2/3. Classical RK4 uses four tendency weights 1/6, 1/3, 1/3 and 1/6. These formal orders assume a consistently evaluated right-hand side. Cadence holding, phase switches and limiter activation can change the accuracy of the coupled model. Nimbus must stage and restore primary fields, private moments and reports consistently if

any stage fails. The time-integration tests and runtime timestep tests check different parts of this requirement.

3.7 A concrete mass–heat–pressure example

Consider an illustrative scalar cell containing 10 kg of dry air and a uniform vapor mixing ratio of 0.01. During transport, 2 kg of dry air enters and 1 kg leaves, carrying the same mixing ratio. Its final dry mass is 11 kg; vapor mass is $0.1 + 0.02 - 0.01 = 0.11$ kg, so its mixing ratio remains 0.01. Using the old 10 kg denominator after updating vapor mass would incorrectly produce 0.011. This elementary example is the reason continuity and scalar transport share one flux history.

Next suppose a liquid-phase source condenses 2×10^{-4} kg per kg dry air. The source removes that amount from vapor and adds it to cloud liquid, so total water is unchanged. Under the ordinary constant-c_{pd} latent-heating convention, its temperature increment is

$$\Delta T_{\text{source}} = \frac{L_v}{c_{pd}} 2 \times 10^{-4} \simeq 0.498 \text{ K}.$$

At held stage pressure, this becomes $\Delta\theta = \Delta T_{\text{source}}/\Pi$ with $\Pi = (p/p_{\text{ref}})^{\kappa_d}$. This example illustrates the units and the phase-source convention; a real saturation solver determines the amount from the coupled saturation/heating calculation rather than accepting an arbitrary condensation amount.

Now isolate the later dry-EOS projection and suppose the initial cell was EOS-consistent, with $T = 280$ K. Hold its post-transport dry density fixed and omit all intervening sources for this illustration. In exact arithmetic the new EOS temperature is

$$T_{\text{EOS}} = T \left(1 + \frac{\Delta T_{\text{source}}}{T}\right)^{\gamma_d}.$$

To first order its increase is $\gamma_d \Delta T_{\text{source}}$, about 0.70 K, rather than 0.498 K. The difference follows from the retained thermal source convention and the subsequent pressure diagnosis; it cannot be removed from the explanation by silently replacing c_p with c_v. Actual float storage adds a further small contribution. In the full runtime, intervening operators also contribute and require their own attribution.

Nimbus's energy accounting records committed source and EOS changes. A closed ledger says those changes have been assigned consistently. Establishing that the complete thermal/mechanical exchange is physically conservative requires the stronger coupled source, EOS, transport and boundary argument. The energy tests and guarded source/EOS tests make that distinction explicit.

3.8 Boundaries complete the equations

A periodic domain cancels opposing boundary exchanges. A closed physical wall imposes its normal-flow constraint and may still have thermal or tangential stress fluxes if the chosen surface model supplies them. An open boundary lets mass, momentum and energy leave or enter. Radiative conditions approximate outgoing disturbances using a finite history; they are neither exact exterior solutions nor permission to reuse stale stage data.

Sponge and Rayleigh terms deliberately relax fields. They can remove or add kinetic and thermal energy according to the target state. Sedimentation removes water through the lower surface, while

radiation transports energy through the top and bottom of a column. Consequently a domain energy curve is meaningful only alongside those exchange terms and the chosen inventory basis. The boundary tests and native C-grid boundary tests exercise these conditions at their actual locations and stages.

Neither a stable integration nor an accounting identity is a complete validation of a storm simulation. Resolution sensitivity, physical source references, supported coupling, restart, backend parity and scientific review remain separate questions. The runtime catalog and repository readiness contract determine which combination can make which claim.

Chapter 4

The grid as a physical and numerical model

A grid does more than select where an equation is evaluated. It decides which volume owns mass, which surface carries a flux, which neighboring states can influence one another, and which motions can be represented. These decisions must agree between dynamics, microphysics, terrain, diagnostics and output. This chapter develops those agreements from simple cases before considering stretched grids and nested domains. The derivations below explain constraints on the implementation; they are not claims that every Nimbus operator is an exact realization of every identity derived here.

4.1 A cell average, a sample and an inventory are different objects

Let C be a physical cell with volume V_C. For a continuous density field, the cell-average dry density is

$$\overline{\rho_d}_C = \frac{1}{V_C}\int_C \rho_d \, dV, \qquad M_C = V_C \overline{\rho_d}_C.$$

If a vapor field varies inside the cell, the dry-mass-weighted mixing ratio is

$$\tilde{q}_{v,C} = \frac{\int_C \rho_d q_v \, dV}{\int_C \rho_d \, dV}, \qquad W_{v,C} = M_C \tilde{q}_{v,C}.$$

In general, this weighted average is not the arithmetic volume average of q_v, and neither average has to equal the value at the geometric center. Writing one floating-point number per cell necessarily chooses a representation of the unresolved variation. Nimbus's conservative scalar transport treats $\rho_d V$ as the carrier inventory and Mq as the transported species inventory. A plotted center value is a reconstruction of that representation.

The distinction matters even in an apparently simple two-cell average. Suppose two equal-volume cells contain dry masses of 2 and 1 kg and have mixing ratios of 0.01 and 0.02. Their total water is 0.04 kg and total dry mass is 3 kg, so the combined mixing ratio is 0.04/3. The arithmetic mean, 0.015, would imply 0.045 kg of water after combining them. No equation of state or more accurate color map can repair that inventory error afterward. Restriction, remapping and budget calculations must preserve the mass weights at the operation that changes the representation.

Nimbus is not uniformly a finite-volume discretization of every prognostic quantity. Scalar continuity and conservative transport have explicit inventory contracts; velocity tendencies, pressure projection and closures have their own discrete forms. The conservation of one scalar update does not establish a conservative total-energy discretization of the complete model. The useful question is which identity an operator satisfies, using which state and which geometry, rather than whether the entire program is given one numerical label.

4.2 Allocated dimensions are not the atmospheric domain

The arrays are contiguous containers with index

$$\ell(i, j, k) = (iN_y + j)N_z + k.$$

The last index is contiguous in memory. This is a storage rule, not a definition of a cell's physical volume or a promise that each index is an independent atmospheric cell. The historical member names `nr` and `nth` are also used by Cartesian paths; their spelling does not make those paths cylindrical. Coordinates, staggering and the declared physical domain supply the interpretation.

PhysicalCellDomain uses half-open ranges, so $i_{\text{begin}} \le i < i_{\text{end}}$ owns scalar control volumes. In the ordinary arrays with at least three points on an axis, one horizontal plane at each side is boundary-closure storage. The collocated layout similarly excludes one plane at each vertical end when there are at least three vertical entries. The C-grid scalar layout retains its vertical scalar layers, with its ground-face velocity handled separately. Tiny arrays used by bounded tests have explicit special cases; do not derive a universal halo width from a single fixture.

For example, an allocated $10 \times 8 \times 6$ scalar array has 480 storage locations. Under the C-grid ownership rule it contains $8 \times 6 \times 6 = 288$ physical scalar cells. Under the ordinary collocated rule it contains $8 \times 6 \times 4 = 192$. These counts describe the current ownership helper, not recommended simulation dimensions. Multiplying 480 by a nominal cell volume would count closure state as additional atmosphere in either example. Likewise, inferring the physical domain length from the allocation alone can change a prescribed wavelength, mean density or column depth.

A halo can contain a finite, plausible value copied from an interior cell. That does not give it volume. It exists so a stencil can read a defined neighbor, so a boundary algorithm can retain history, or so a restart can reproduce the same closure state. A valid restart may therefore retain more data than a physical inventory integrates. Output metadata must carry the ownership range rather than asking a reader to guess it from zeros, NaNs or array dimensions.

4.3 Three distances that must not be collapsed into one

Consider a flat but vertically stretched column. Let the lower and upper interfaces of layer k be $z_{k-1/2}$ and $z_{k+1/2}$, and let its representative scalar height be z_k. The layer thickness is

$$h_k = z_{k+1/2} - z_{k-1/2},$$

whereas the distance used by a center-to-center pressure gradient is $d_{k+1/2} = z_{k+1} - z_k$. A vertical-velocity dual control volume spanning two adjacent scalar centers has a width related to $d_{k+1/2}$, not automatically to either neighboring h_k. All three distances coincide in a special uniform construction; that coincidence is the reason the distinction can disappear unnoticed in a uniform-grid implementation.

For a piecewise linear reconstruction between two scalar locations, the value at an intermediate interface z_f is

$$\phi_f = \frac{z_{k+1} - z_f}{z_{k+1} - z_k}\phi_k + \frac{z_f - z_k}{z_{k+1} - z_k}\phi_{k+1}.$$

A simple half-sum is justified only when the face is halfway between those locations. An independently reconstructed upwind value is a different numerical choice. Neither should be substituted for the other just because both functions return a value at an interface.

The divergence of a vertical flux density f in layer k is $(f_{k+1/2} - f_{k-1/2})/h_k$. The gradient at an interior interface instead uses $(\phi_{k+1} - \phi_k)/d_{k+1/2}$. Substituting one global Δz into both expressions changes the amount of transport per layer and the force between layers. It can also destroy the cancellation between a base pressure gradient and gravity while leaving the fields visually smooth.

The physical metric interface validates scalar centers, interfaces and positive layer thicknesses and supplies the geometry used by column physics. It rejects a partly initialized flat geometry and a terrain geometry whose shape does not match the fields. Radiation, sedimentation, vertical mixing and diagnostics must consume that same physical column. A radiation heating rate based on a nominal layer depth while transport uses the actual depth is a coupling inconsistency, not merely a small plotting discrepancy.

4.4 Why C-grid staggering changes resolved waves

The C-grid locates a normal velocity at the face through which it transports mass. In Nimbus's index convention, $u(i, j, k)$ is the east face of scalar cell (i, j, k), $v(i, j, k)$ its north face and $w(i, j, k)$ its upper face. Thus the west face used by a cell is addressed with the neighboring index. The separately owned lower ground face is an important endpoint of the vertical construction. A physical scalar interval of n cells has $n + 1$ boundary-and-interior face locations in a nonperiodic direction. Under periodic conditions the end faces represent one physical connection and must not become two distinct exchanges in an inventory sum.

The benefit can be seen without a storm simulation. For a uniform one-dimensional pressure perturbation $p_j = \hat{p}\exp(\mathrm{i}kj\Delta x)$, a collocated centered gradient has the Fourier factor

$$\frac{p_{j+1} - p_{j-1}}{2\Delta x} = \mathrm{i}\frac{\sin(k\Delta x)}{\Delta x}p_j.$$

It is blind to the alternating grid-scale pattern at $k\Delta x = \pi$. A center-to-face C-grid difference instead has magnitude $2\sin(k\Delta x/2)/\Delta x$ after accounting for the half-cell phase shift. It does not vanish at that wavelength. This explains one pressure–velocity coupling advantage of staggering. It does not imply that a two-cell wave is accurate or that an A-grid implementation cannot provide a different consistent coupling with its own stabilization and reconstruction.

For the illustrative linear acoustic pair $p_t = -\rho_0 c^2 Du$ and $u_t = -Gp/\rho_0$, a compatible C-grid divergence and gradient on a periodic uniform grid give

$$\omega_h^2 = c^2\left[\frac{2\sin(k\Delta x/2)}{\Delta x}\right]^2.$$

The continuous relation is $\omega^2 = c^2k^2$. Their difference is a spatial dispersion error before the time integrator contributes any additional error. At small $k\Delta x$, expanding the sine shows convergence to the continuous wave; near the shortest represented wavelength, the discrete phase and group

velocities differ substantially. Increasing the formal order of an unrelated microphysics source does not repair that wave dispersion.

A nominal grid spacing therefore does not equal the smallest accurately resolved physical structure. The useful resolved range depends on the complete spatial operator, filtering, time step, closures and the diagnostic being measured. A resolution study must compare that diagnostic across grids under consistent physical assumptions. Holding a number of grid cells fixed while changing Δx changes the physical size of a perturbation; holding a Smagorinsky coefficient fixed while changing its grid-derived filter width changes the modeled subgrid diffusivity.

4.5 Flux cancellation is a topology statement

Assign each interior face one orientation and one numerical dry-mass flow F_f. Let B_{Cf} be $+1$ when that orientation points outward from cell C, -1 when it points inward, and zero otherwise. The update is

$$M_C^{n+1} = M_C^n - \Delta t \sum_f B_{Cf} F_f.$$

An interior column of B contains one $+1$ and one -1. Summing over all physical cells cancels that face exactly in real arithmetic, regardless of whether the reconstruction is first or fifth order. What remains is the physical boundary flux. This is the fundamental topological reason shared fluxes conserve an inventory.

If the two adjacent cells separately reconstruct and round what is supposed to be the same interface flow, they can use $F_f + \epsilon_L$ and $F_f + \epsilon_R$. The domain acquires a residual proportional to $\epsilon_L - \epsilon_R$. Thread safety alone does not establish a shared flux: two race-free calculations can still disagree. Conversely, deterministic summing of a wrong set of volumes does not make an inventory physical.

For a mixing ratio, Nimbus's retained continuity flux provides the same carrier flow to the species update,

$$Q_C^{n+1} = Q_C^n - \Delta t \sum_f B_{Cf} F_f q_f, \qquad q_C^{n+1} = Q_C^{n+1} / M_C^{n+1}.$$

When $q_f = q_0$ everywhere, the uniform value survives exactly in real arithmetic: $Q_C^{n+1} = q_0 M_C^{n+1}$. This is stronger than obtaining the correct domain sum alone. A scheme can preserve total water yet generate false cell-scale concentration changes if it transports a different dry-mass carrier. The retained-flux transport contracts test this consistency along with positivity and invalid-input behavior.

The algebra also exposes a time-level requirement. If Q_C^n is constructed from an already updated density and an old mixing ratio, it is not the old inventory. Saving the old carrier, or constructing the inventory before the carrier is overwritten, is part of the mathematical scheme. Eliminating that snapshot for performance requires an equivalent ownership arrangement. It is not safe merely because no API signature changes.

4.6 Reconstruction determines accuracy; limiting determines admissibility

A face flow needs a transported value. In a positive one-dimensional flow, a first-order upwind method takes the value from the cell on the inflow side. For constant velocity $a > 0$ and uniform

spacing, the update is

$$q_j^{n+1} = (1 - C)q_j^n + Cq_{j-1}^n, \qquad C = a\Delta t/\Delta x.$$

For $0 \leq C \leq 1$, the new value is a convex combination and cannot exceed the two contributing extrema. This explanatory result assumes a constant carrier, no source and the stated first-order discretization. It is not a universal positivity proof for a compressible, multidimensional, source-coupled update.

A linear reconstruction uses a slope to represent subcell variation and reconstruct left and right interface states. An unlimited centered slope is accurate on a smooth field but may produce negative condensate next to a sharp edge. A limiter reduces the slope when the neighboring differences suggest an extremum or discontinuity. WENO methods instead combine candidate reconstructions with smoothness-dependent weights. Their motivation is to recover high order in smooth regions while avoiding an oscillatory stencil near a sharp feature. The reconstruction, its limiting, its boundary stencil and its time integration must be evaluated together.

Nimbus's numerical implementations and the retained numerical tests define the admitted transport paths. The method name does not establish high-order convergence everywhere: a limiter can intentionally revert toward lower order, a boundary may use a shorter stencil, and a phase source can introduce nonsmooth behavior. A resolution comparison should report where the observed error comes from, rather than assigning the formal smooth-field order to an entire convective storm.

Clipping a negative mixing ratio after a flux update changes its inventory. If a stored cell would contain $Mq = -\delta$ and is simply set to zero, the operation adds δ to the domain water amount. Small clipping is therefore not automatically harmless. A conservative correction must identify where the corresponding amount is removed, respect the available donor inventory, and report the adjustment. A physical source limiter has a related duty: multiple processes cannot each consume the same full reservoir independently.

4.7 Pressure work needs compatible scalar and velocity geometry

On a uniform grid with periodic boundaries, summation by parts gives a discrete counterpart of the continuous pressure-work identity. For an interior face between left and right cells with area A_f and center distance d_f,

$$\sum_C V_C p_C (Du)_C = \sum_f A_f u_f (p_L - p_R) = -\sum_f A_f d_f u_f (Gp)_f.$$

The velocity dual volume is $V_f^* = A_f d_f$. At a physical boundary there is an additional boundary pressure-work term. This identity depends on using the same face orientation, areas and center distances on both sides. Interpolating a pressure gradient with one geometry and diagnosing work with another breaks the pairing.

If the velocity equation uses $\rho_f \, du_f/dt = -(Gp)_f$, multiplying by $V_f^* u_f$ supplies the kinetic-energy pressure work. Its partner in the internal energy equation has the opposite sign. This is a design criterion for a coupled energy discretization. Nimbus's diagnosed energy budgets and pressure/EOS operators must be assessed against their actual discrete contracts; the continuous identity by itself does not prove exact total-energy conservation for the implemented rho–theta model.

A staggered kinetic inventory must first decide which cell mass weights its velocity locations. As an illustrative dry inventory, retain the dry mass M_C defined above. On a flat grid, assigning half

of each adjacent cell's mass gives

$$M_f^* = \tfrac{1}{2}M_L + \tfrac{1}{2}M_R, \qquad K_f = \tfrac{1}{2}M_f^* u_f^2.$$

Boundary faces, vertical stretching and mapped ground motion require the appropriate half-volume and metric construction. Averaging velocity to scalar centers and then squaring is a different functional because the square of an average is not the average of squares. Nimbus's moist-mixture basis-v2 diagnostic instead uses cell masses $M_C^{\mathrm{mix}} = \rho_{d,C} V_C (1 + q_{t,C})$ in this half-cell construction. That diagnostic choice does not change the dynamical pressure-force convention, which uses its declared dry-density approximation. The flat-grid velocity/energy pairing implementation and the full energy diagnostic make the mass basis and geometric scope explicit.

Time discretization adds another issue. At fixed face mass, an impulse I changes velocity by I/M_f^* and changes kinetic energy by

$$\Delta K_f = u_f^n I + \frac{I^2}{2M_f^*}.$$

The second term is missing if work is calculated using only the old velocity. When face mass also changes, a term for that mass change is required. For example, the exact endpoint decomposition

$$\Delta K_f = \tfrac{1}{2}M_f^{*,n+1}\left[(u_f^{n+1})^2 - (u_f^n)^2\right] + \tfrac{1}{2}(M_f^{*,n+1} - M_f^{*,n})(u_f^n)^2$$

is algebraically valid but assigns work differently from a midpoint-based construction. An energy report must declare its endpoint basis; combining terms from two different decompositions can double-count a contribution.

4.8 Terrain is a coordinate transformation, not a tilted picture

For the explanatory mapping $(x, y, \zeta) \mapsto (x, y, z(x, y, \zeta))$, define $J = z_\zeta > 0$, $z_x = (\partial z/\partial x)_\zeta$ and $z_y = (\partial z/\partial y)_\zeta$. Chain rules give

$$\begin{aligned}
(\partial_x \phi)_z &= (\partial_x \phi)_\zeta - \frac{z_x}{J}\partial_\zeta \phi, \\
(\partial_y \phi)_z &= (\partial_y \phi)_\zeta - \frac{z_y}{J}\partial_\zeta \phi, \\
\partial_z \phi &= \frac{1}{J}\partial_\zeta \phi.
\end{aligned}$$

Nimbus's mapped transport interface uses $G_x = -z_x$ and $G_y = -z_y$ with this Jacobian convention. The signs are part of the interface. Reusing a formula from a paper whose metric symbols have the opposite sign without converting it can reverse the terrain correction while preserving dimensions.

For a stationary mapping the contravariant vertical coordinate velocity is

$$\dot{\zeta} = \frac{w - u z_x - v z_y}{J}.$$

The transformed dry-mass equation can be written

$$\partial_t(J\rho_d) + \partial_x(J\rho_d u) + \partial_y(J\rho_d v) + \partial_\zeta\{\rho_d(w - u z_x - v z_y)\} = 0.$$

This equation is written for the stated mapping and continuous metrics. A finite-volume implementation must still construct compatible discrete face metrics, fluxes and physical volumes. It

cannot obtain that compatibility merely by evaluating the continuous expressions independently at arbitrary points.

At stationary terrain $z = h(x, y)$, zero normal flow requires $w = uh_x + vh_y$. Setting the Cartesian vertical velocity to zero instead imposes a different condition unless the terrain is locally flat or the wind has no component along the slope gradient. A column's lower interface can therefore have a physical vertical-velocity contribution even when no mass crosses the ground. The distinction carries into surface stress, terrain pressure work and the staggered kinetic inventory.

A useful geometric consistency check is free-stream preservation. A constant physical velocity field should not acquire spurious divergence solely because coordinate surfaces slope. The discrete metric construction must allow the corresponding face contributions to cancel. A second check is hydrostatic balance: pressure and gravity should cancel for the declared resting base state using the same physical heights and averaging. These are different tests. Passing one does not prove the other or establish the accuracy of transport through a sharp terrain transition.

See the mapped gradient and transport contract, its implementation, and mapped reconstruction checks. The WRF technical notes provide a useful comparison with a mature terrain-following solver. WRF's vertical coordinate and complete equation set are not Nimbus's; that reference motivates questions about metrics and staggering rather than supplying an interchangeable formula set.

4.9 A moving window is not a new physical wind

For a moving physical control volume, Reynolds transport uses relative normal velocity $(u-v_g)\cdot n$. If the window translates horizontally with velocity U_g, a stationary Earth-frame parcel flows backward through that window even though its physical wind is zero. Stored Earth-frame momentum must not be incremented by a window shift. The relative transport velocity and the physical velocity used for kinetic energy have different purposes.

A continuously deforming coordinate system also obeys the geometric conservation law: the change in its volume must match the surface integral of grid velocity. That is a constraint on a general moving-grid discretization, not a claim that Nimbus implements unrestricted mesh deformation. Nimbus's domain controls, integer relocation, terrain placement and supported nesting profiles have specific execution and admission contracts. A discrete relocation is a remap operation with retained and newly exposed regions; it is not equivalent to integrating a continuous ALE flux over an arbitrary deformation.

To understand a conservative remap, let m_{ij} be dry mass transferred from old cell j into new cell i. The transferred species amount is $m_{ij}q_j$ in a piecewise-constant example. Then

$$M_i^{\text{new}} = \sum_j m_{ij}, \qquad Q_i^{\text{new}} = \sum_j m_{ij}q_j, \qquad q_i^{\text{new}} = Q_i^{\text{new}}/M_i^{\text{new}}.$$

If the transfers are nonnegative, the reconstructed ratio is bounded by its donors. If $\sum_i m_{ij} = M_j^{\text{old}}$ for every fully retained old cell, the remap conserves its mass. Cells leaving a limited window and fresh cells entering it require explicit departure and inflow accounting. The mass inside a tracking window need not remain constant even when the remap is correct.

Private particle moments, held boundary-layer sources, turbulence state and boundary history also have location and time ownership. Preserving the visible q arrays while dropping a number moment changes the size distribution. Preserving a held heating array while assigning it to the wrong Earth

location changes the forcing. The domain-recycling implementation and its provenance records identify the geometric ownership that a relocation must preserve.

4.10 Nesting introduces an interface budget

A fine child overlaps part of a coarse parent. Adding every parent cell and every child cell counts the overlap twice. A composite inventory instead includes the uncovered parent plus the child representation of the covered region. This choice must be explicit before asking whether two-way coupling conserves mass or energy. A domain total from one grid and a composite total are different diagnostics even if they use the same units.

For restriction of a conserved scalar, the coarse covered inventory is the sum of its contributing child inventories. The coarse mixing ratio follows by dividing summed species mass by summed dry mass. Face-normal momentum and staggered velocity require their own geometry and cannot be restricted by copying the scalar formula blindly. Prolongation must likewise distinguish preserving a parent mean from creating artificial fine-grid extrema.

Flux mismatch is a separate concern. Suppose the coarse update removes $\Delta t F_c$ at a coarse/fine interface, while fine substeps remove a total $\sum_s \Delta t_s \sum_{f \subset c} F_{f,s}$. A correction replacing the coarse exchange with the fine exchange has the illustrative signed form

$$\delta M_c = \Delta t F_c - \sum_s \Delta t_s \sum_{f \subset c} F_{f,s}.$$

The sign reverses with the orientation convention. Matching instantaneous fluxes at the final time is not enough when the child subcycles. Water and thermal carriers also need consistent exchanges; matching dry mass alone does not establish the coupled conservation result.

Nimbus separates nested level execution, scalar feedback, terrain geometry and boundary ownership. The feedback contract, feedback implementation and level execution are the relevant source boundaries. The interface arguments above explain what must be checked; they do not assert that every nested configuration implements this particular reflux formula or that arbitrary combinations are admitted. The public runtime catalog remains authoritative about the supported combination.

4.11 Spatial resolution also changes the time and closure problem

An advective time scale in a cell is set by its outgoing relative flux compared with its carrier inventory. A useful dimensional measure is

$$C_M = \frac{\Delta t}{M_C} \sum_{f\,\mathrm{out}} |F_f|.$$

For simple upwind transport this is directly related to positivity. The exact stability bound depends on the spatial operator and time integrator. Acoustic steps have the additional sound-speed scale; explicit diffusion has a scale proportional to h^2/K; sedimentation has a fall-speed crossing time. Improving one constraint does not remove the others.

On a uniform Cartesian mesh, familiar explanatory estimates are

$$\Delta t_{\mathrm{adv}} \sim \frac{h}{U}, \qquad \Delta t_{\mathrm{acoustic}} \sim \frac{h}{c+U}, \qquad \Delta t_{\mathrm{diff}} \sim \frac{h^2}{K}.$$

The constants and directional combinations must come from the implemented scheme, not these dimensional estimates. Mapped grids must use the relevant physical metric; a single nominal horizontal spacing does not detect a very thin vertical layer. Nimbus exposes acoustic substep selection and time-step history through the time-stepping contract, with separate stability and cadence owners elsewhere in the runtime.

Refining all three spatial axes by two creates roughly eight times as many physical cells. If a wave or advective constraint also halves the time step, work per simulated interval grows by about sixteen before changes in solver iterations, output, memory traffic or physics subcycling. This is a scaling argument, not a benchmark prediction. If explicit diffusion becomes limiting, the time-step penalty can be larger. Refinement also changes grid-derived SGS filter widths and the relation between resolved clouds and bulk closures. A scientifically useful convergence experiment must specify which physical parameters stay fixed and which model assumptions change with resolution.

Chapter 5

Bulk microphysics, particle moments and thermal coupling

Microphysics connects resolved ascent, water vapor, cloud particles and precipitation. A resolved updraft changes density and temperature, which changes saturation. Condensation then changes phase and heats the air; particles collide, freeze or melt; precipitation falls relative to the air and can evaporate into a downdraft. These processes feed back on pressure, buoyancy and condensate loading. Their numerical order and units are consequently as important as the formula for any isolated rate.

This chapter explains the implemented scheme families and their assumptions. It supplements the mathematics overview and the grid and dynamics chapter. The executable's catalog and runtime admission define supported combinations. In particular, the presence of internal triple-moment source code does not make public triple-moment Milbrandt selection available.

5.1 What a bulk category represents

A bulk category collects particles with related physical behavior into one population. Cloud droplets and rain are both liquid, but their characteristic sizes, collision behavior and fall speeds differ. Cloud ice, snow, graupel and hail likewise represent different frozen populations. A conversion between categories is not necessarily a change of phase: collecting rain into larger rain drops releases no fusion or vaporization heat, and aggregating ice into snow does not melt it.

Nimbus stores category mass as $q_x = m_x/m_d$, in kg per kg dry air. Multiplying by dry density gives water content per physical volume, $\rho_d q_x$. This is the quantity required by many published collision and sedimentation kernels. Private number fields similarly declare whether they are per dry mass or per volume. A missing density conversion changes both the size distribution and the collision frequency.

A single-moment family predicts category mass and diagnoses other properties from an assumed spectrum. A double-moment population also predicts number, so the mean particle mass q/N can vary independently of the total mass. A third independent moment constrains another property of the distribution, but does not resolve its arbitrary shape. These labels describe closure information; they are not an accuracy ranking valid for every cloud or process.

5.2 Water transfers and latent heat

Consider a positive accepted transfer amount C over a source interval. A conservative vapor-to-cloud transfer has

$$\begin{aligned} q_v' &= q_v - C, \\ q_c' &= q_c + C, \\ q_t' &= q_t. \end{aligned}$$

In the ordinary constant-heat-capacity source convention, the associated temperature change is $L_v C/c_{pd}$. Freezing liquid releases $L_f C/c_{pd}$; deposition from vapor to ice releases $L_s C/c_{pd}$, where $L_s = L_v + L_f$. Evaporation, melting and sublimation reverse those signs. Liquid-to-liquid and frozen-to-frozen transfers have no latent phase-change term.

At the pressure used to evaluate the source, the temperature tendency converts to potential temperature through

$$\begin{aligned} \Pi &= (p/p_{\mathrm{ref}})^{R_d/c_{pd}}, \\ \dot{\theta}_{\mathrm{source}} &= \dot{T}_{\mathrm{source}}/\Pi. \end{aligned}$$

There is no additional division by T in this expression. Such a factor would belong to a logarithmic thermal variable, which is not what the field stores. This fixed-stage-pressure convention is also not a derivation of the thermal state after the later dry-EOS projection. The last sections below separate the ordinary source convention from the guarded coupled endpoint owner.

An unconstrained rate $\dot{C}$ becomes an interval amount $\Delta t \dot{C}$. The amount must fit its donor, and competing processes must have a consistent claim on that donor. Ordinary schemes can use sequential working populations: material produced early in their declared source order can be consumed later. The guarded wet-hail composition instead records original donors before returns and forbids selected same-interval reuse. Neither policy is universal; each defines a finite-step approximation that must be tested as implemented.

For a physically closed cell-source calculation, species increments should sum to zero before storage rounding. If an implementation first overdraws a donor and then clips the negative result, positive products need not be removed with it. A final nonnegativity clamp therefore does not establish water conservation or the correct latent heat. Nimbus's maintained checks examine the accepted transfers and their thermal changes, as well as invalid-input and rollback behavior where the owner exposes a transaction.

5.3 Saturation is coupled to the heat it releases

For an ideal vapor mixture, the liquid saturation mixing ratio is represented by a relation of the form

$$q_{vs}(T,p) = \epsilon \frac{e_s(T)}{p - e_s(T)}, \qquad \epsilon \simeq R_d/R_v.$$

Nimbus uses retained empirical water and ice vapor-pressure formulas and bounded temperature, pressure and saturation domains in its shared thermodynamic helpers. The water and ice saturation

values differ in a supercooled cloud. Air can be subsaturated relative to liquid and supersaturated relative to ice, so an ice deposition process must not be gated solely by liquid relative humidity. The helper bounds are numerical/model domains, not evidence that the empirical fit is physically valid for arbitrary atmospheric conditions.

Condensing all of the initial excess $q_v - q_{vs}(T, p)$ while leaving temperature unchanged would overestimate condensation: latent warming raises saturation. Under the shared liquid-adjustment convention, the coupled problem is

$$\begin{aligned} q_v' &= q_{vs}(T', p), \\ T' &= T + \frac{L_v}{c_{pd}}(q_v - q_v'), \\ q_c' &= q_c + q_v - q_v'. \end{aligned}$$

It has a finite-reservoir alternative. If all existing cloud evaporates before saturation is reached, the endpoint has $q_c' = 0$ and remains subsaturated, with cooling from the accepted evaporation. Saturation is not enforced by creating cloud water of the wrong sign.

Linearizing the saturation response gives an instructive approximation:

$$C \simeq \frac{q_v - q_{vs}(T, p)}{1 + (L_v/c_{pd})\,\partial_T q_{vs}(T, p)}.$$

Using a Clausius–Clapeyron slope approximation gives the familiar denominator

$$1 + \frac{L_v^2 q_{vs}}{c_{pd} R_v T^2}.$$

This explains the thermal feedback factor; it does not make the linearized solve identical to a nonlinear saturation root. Nimbus's shared liquid adjustment uses a bounded safeguarded nonlinear iteration, with a fixed iteration limit and explicit finite-cloud handling. The retained Lin source instead uses its own linearized adjustment. Their differences are part of the scheme definitions.

5.4 Kessler: a liquid-only warm-rain model

Nimbus's Kessler family evolves vapor, cloud liquid and rain. Frozen categories in the common interface do not participate in its sources or sedimentation. This is a purposeful reduced model for liquid processes; it cannot explain mixed-phase precipitation by implicitly treating an ignored frozen field as another rain reservoir.

The source first applies the shared liquid saturation adjustment. Warm-rain production then combines threshold autoconversion with rain accretion. Autoconversion represents formation of precipitating drops from cloud droplets without resolving their collision spectrum; accretion represents existing rain collecting cloud. In the retained finite-step Kessler/COMMAS form, define

$$\begin{aligned} f &= \left(1 + c_{\mathrm{accr}} \Delta t \, \max(q_r, 0)^{0.875}\right)^{-1}, \\ P_* &= q_c(1 - f) + f \, c_{\mathrm{auto}} \Delta t \, \max(q_c - q_{c0}, 0), \\ P &= \min(q_c, \max(0, P_*)). \end{aligned}$$

The source subtracts P from cloud and adds it to rain. The factor f is a finite-step implicit treatment of the cloud sink at the retained rain amount; the threshold term turns off autoconversion below q_{c0}. Since both species are liquid, this transfer has no latent heating.

Rain evaporation uses the retained diffusion/ventilation kernel. Its available rain includes rain just produced by the preceding conversion. The accepted evaporation is bounded by that rain reservoir and the remaining thermally adjusted saturation deficit. The ventilation factor models enhanced vapor exchange around falling drops; it does not resolve their surrounding flow. Rain falls with the prescribed terminal-speed law, density correction and configured speed bound.

The coefficient names and units are part of the retained parameterization. Replacing the $q_r^{0.875}$ law with a visually similar linear expression, or forbidding newly produced rain from evaporating, would alter the model. See Kessler's implementation and the reference-column tests.

5.5 Lin: sequential single-moment mixed-phase transfers

The Lin family adds cloud ice, snow, graupel and hail to the liquid system. Nimbus retains a Lin-type process implementation inspired by Lin, Farley and Orville (1983), with explicit local choices and simplifications. The family name does not mean that every rate is a literal reproduction of the complete paper.

The pointwise source creates working copies of all seven water mixing ratios. Each transfer takes mass from its current working donor, adds the same amount to its destination and accumulates the appropriate latent temperature change. The process temperatures and saturation quantities are based on the entry thermodynamic state; the masses evolve sequentially through the source order. The retained temperature workspace includes a float rounding step for CPU/native consistency. Spectrum-integral density is diagnosed as $p/(R_d T)$ in this path.

The warm portion uses the linearized liquid saturation adjustment, threshold cloud autoconversion, cloud–rain accretion and relative-humidity-dependent rain evaporation. Below freezing, cloud freezing feeds cloud ice, stochastic rain freezing feeds hail embryos, and vapor deposition grows ice. Ice collecting cloud contributes graupel, while light snow riming grows snow. Ice-to-snow aggregation uses a temperature-dependent threshold law and no latent phase heat. Ice and snow can sublimate when subsaturated relative to ice. In warm air, the retained snow, graupel and hail melting terms feed rain with a fusion-heat sink. Rain and precipitating frozen species also sediment.

These categories carry mass without a separately transported number family. The assumed particle properties used in integral rates therefore matter: changing an assumed intercept or density changes collection and fall speed even when category mass is unchanged. The cloud-freezing ramp, routing of frozen drops, and distinct melting coefficients are implementation choices visible in the Lin source. Reference-process tests check those concrete transfers and their latent heat, rather than treating agreement of a final rainfall total as sufficient verification.

5.6 Thompson: mixed moments, aerosols and tabulated processes

Nimbus's Thompson family carries vapor, cloud liquid, rain, cloud ice, snow and graupel. Nonzero hail input is rejected by this implementation. Its private prognostic state has five number fields: cloud number, rain number, ice number, water-friendly aerosol and ice-friendly aerosol. Their transport bindings are per kg dry air. This is more specific than describing the family as uniformly single- or double-moment: different populations have different closures.

Activation turns eligible aerosol into cloud droplets under the parameterized supersaturation/updraft environment; ice nucleation supplies ice through its own aerosol and thermodynamic assumptions. Collision and scavenging change particle and aerosol populations. The resulting number affects mean mass and therefore fall speeds and subsequent collision integrals. Aerosol profile choices are scientific initial conditions, and portable restart must retain the number state rather than silently regenerate it from a default profile. See the private-state owner and its transport/restart tests.

Thompson et al. (2008) emphasize a snow distribution that depends on temperature and ice-water content, including non-spherical mass–size behavior. Nimbus's corresponding distribution code diagnoses snow properties from snow water content, temperature and air density; snow number is not an additional transported field in this owner. Its retained snow mass law uses $m(D) = 0.069D^2$ in SI units, so the second diameter moment is snow water content divided by 0.069. Empirical moment relations and the two-component distribution determine the other integrals. This differs from assuming constant-density spherical snow, for which mass would scale as D^3.

The implementation has separate owners for activation, nucleation, warm collection and evaporation, cold deposition/sublimation, freezing, collection, melting, sedimentation and terminal cloud-phase completion. Number-weighted and mass-weighted velocities differ where the retained spectrum supports them. Such differences allow size sorting: the population's mass and its count need not fall at one identical speed. The particle distributions and distribution tests make the moment and unit conventions explicit.

Several processes use packaged activation, collection and cold-process tables. These are required scientific inputs with pinned identities, not disposable benchmark artifacts or optional caches. Generators and reference tests retain their upstream source correspondence; runtime loads the packaged tables and does not call live Fortran. Missing or corrupt required data fail rather than switch to an unrelated approximation. The Thompson source guide, collection tests and cold-process tests describe the retained scope. Neither the table lineage nor the scheme name alone establishes that every feature of every Thompson publication is present.

5.7 Milbrandt: use moments to determine a spectrum

Ordinary Milbrandt transports number for cloud, rain, ice, snow, graupel and hail, in addition to the primary mass fields. Each number is per kg dry air. The mass and number constrain a characteristic size; a prescribed or diagnosed shape relation supplies the remaining information needed by a double-moment distribution. The ordinary public path does not prognose an independent sixth diameter moment.

For clarity, write a gamma spectrum per kg dry air as

$$\begin{aligned} n(D) &= N_0 D^{\alpha} e^{-\lambda D}, \\ M_k &= \int_0^{\infty} D^k n(D)\, dD = N_0 \frac{\Gamma(\alpha + k + 1)}{\lambda^{\alpha+k+1}}, \\ N &= M_0, \qquad q = cM_d \quad \text{when } m(D) = cD^d. \end{aligned}$$

The moment integral follows by substituting $x = \lambda D$ into the gamma integral. For a given valid shape α, divide mass by number to eliminate N_0. The resulting slope is

$$\lambda = \left[\frac{cN}{q}\frac{\Gamma(\alpha+d+1)}{\Gamma(\alpha+1)}\right]^{1/d}.$$

Increasing mass at fixed number increases typical size. Increasing number at fixed mass makes particles smaller. This difference matters because collection cross-sections scale roughly with squared diameter while fall speed and mass use different powers. The mean-mass diameter is $(q/(cN))^{1/d}$; it is not in general the number-mean diameter $(\alpha+1)/\lambda$.

The retained snow law has coefficient 0.1597 and exponent 2.078 in SI units. Its non-cubic exponent must remain in both inversion and transfer formulas. Replacing it by the cubic exponent used for another species changes the spectrum even if mass and number are unchanged. The law is also distinct from the Thompson snow law described above. See Milbrandt's distribution helpers and sedimentation parameters.

For a pure power-law terminal speed $v(D) = aD^b$, a moment-weighted mean follows directly from the same integral:

$$\overline{v}_k = \frac{\int v(D)D^k n(D)\,dD}{M_k} = a\lambda^{-b}\frac{\Gamma(\alpha+k+b+1)}{\Gamma(\alpha+k+1)}.$$

Number weighting uses $k = 0$, mass weighting uses $k = d$, and sixth-moment weighting uses $k = 6$. Actual kernels also retain their species density factors, speed bounds or additional functional dependence. This derivation explains why one universal velocity cannot transport all moments consistently; it is not a replacement for those species-specific details.

Collision rates likewise integrate a geometric cross-section, relative fall speed and efficiency over the two spectra. Schematically, a collision count contains an integral of $E(D_x, D_y)\,\pi(D_x + D_y)^2|v_x - v_y|/4$ times the two number spectra, with the required volume/dry-mass conversion. Multiplying by the transferred mass yields a mass rate. Self-collection can reduce number without changing category mass; cross-category collection can move mass and change number in different ways. The choice of destination population is therefore a closure decision, not an automatic consequence of water conservation.

Nimbus retains its existing destination rules and number caps. In the relevant rain/frozen collision route, forming new hail requires the enabled hail option, rain mean-mass diameter greater than 2.5 mm and the retained combined-density criterion of at least 650 kg m^{-3}. These are Nimbus choices in the cold-process owner, and are not silently replaced by a literal alternative from a publication. The ordinary source includes cloud adjustment, activation, warm and cold processes and sedimentation; private number publication must agree with the accepted primary update.

5.8 Sedimentation is a boundary flux, not disappearance

Terminal fall speed is particle motion relative to the air. With upward z positive, a species falling at positive downward speed V_x has sedimentation mass flux $-\rho_d q_x V_x$. For a flat layer, its finite-volume inventory update is the difference between incoming flux from above and outgoing flux below. The bottom outgoing flux becomes surface precipitation. It must remain in the domain water budget even though it is no longer stored in atmospheric cells.

This treatment assumes terminal-speed adjustment is sufficiently rapid that particle momentum need not be a separately resolved state. It also assumes the chosen bulk distribution is sufficient

to diagnose weighted speeds. Subcycling limits the distance traveled per source/sedimentation interval and allows donor bounds to be respected; it does not remove the terminal-speed assumption. Layer thickness and density must be physical quantities, especially on stretched or terrain-following columns.

When multiple moments sediment, their inventories and velocities must use their declared units independently. Using the mass velocity for number merely because it preserves one mass budget would suppress size sorting. Conversely, separate moment fluxes can drive an endpoint outside the realizable distribution domain, so a mass-positive result alone is insufficient. The guarded triple sedimentation tests check this additional constraint alongside fallout and layer accounting.

5.9 What the guarded third moment adds

The internal third-moment work uses $Z = M_6$, a sixth diameter moment per kg dry air in the declared storage convention. It is an additional prognostic moment, not a statement that the exponent is three. Its units and density conversion must be distinguished from a volume radar reflectivity factor. A radar forward operator also needs assumptions about particle phase, dielectric behavior and scattering; a field called reflectivity is not interchangeable with every radar product.

Milbrandt and Yau (2005, Part II) provide the moment framework used by this work. At fixed shape and mass law, eliminating λ gives $Z \propto q^{6/d} N^{1-6/d}$. Logarithmic differentiation yields the process relation

$$\frac{\dot{Z}}{Z} = \frac{6}{d}\frac{\dot{q}}{q} + \left(1 - \frac{6}{d}\right)\frac{\dot{N}}{N}.$$

This relation explains how an admitted mass/number tendency can constrain a sixth-moment tendency under that process assumption. It does not guarantee that an arbitrary Euler increment stays positive or realizable. Initiation, conversion, collection and differential sedimentation need their own moment rules; a population with zero mass cannot support arbitrary positive number and sixth moment. The internal moment inversion and independent tests check the corresponding domains.

Public triple-moment selection remains **Engineering incomplete** and guarded. Its bounded owners preserve ordinary double-moment equations, precision and thresholds. Completing one transfer or passing one cell test does not complete all source families, field transport, restart and live source-to-EOS coupling.

5.10 Wet hail: declare the surface and material assumptions

Wet growth has a thermal constraint in addition to a collision rate. A hail surface at the melting point can freeze collected liquid only as rapidly as its heat balance permits. Excess collected liquid must remain liquid or leave the particle. A unit collection efficiency cannot continue to describe a wet surface after all available liquid is frozen or evaporated.

The guarded Nimbus owner assumes a zero-storage interface at $T_r = 273.15$ K. It does not prognose a hail-core temperature or retained film temperature. Collected liquid and frozen material, and original hail selected for melting, start at the ambient cell temperature. Surface processing brings the relevant material to T_r; products are subsequently represented at the cell endpoint temperature. This is an explicit material-temperature path, not a resolved conduction model inside a hailstone.

Let H be signed conductive heat into the interface, C signed vapor condensation, L collected liquid, F collected frozen mass, R liquid frozen at the surface and M original hail melted. All masses are interval amounts per kg dry air; heat is J per kg dry air. With $\tau = T - T_r$, the retained surface balance is

$$H + L_v C + c_l \tau L + c_i \tau F + L_f R = \left(L_f - c_i \tau\right) M.$$

For cold air, $\tau < 0$, so warming incoming liquid or solid to the interface consumes heat: both collected sensible terms are negative. Melting original cold hail requires its warming plus fusion energy, giving the denominator $L_f - c_i \tau$. Freezing liquid releases fusion heat. This sign structure explains why the collected-solid warming term cannot be omitted just because both its entry and destination category are frozen.

Signed surface evaporation is allowed only from admitted liquid; negative surface vapor return cannot finance another simultaneous vapor consumer. The owner preserves original donor masses and moments before returns. Newly transferred snow or hail cannot immediately fund another admitted collection or original-hail melt in that interval. The endpoint must retain positive liquid for the wet branch and remain inside its thermal and spectral domains. These checks apply to ideal and stored source endpoints and ideal and stored post-EOS endpoints. Crossing an unowned wet/dry or cold/warm boundary rejects the interval rather than continuing a rate law outside its domain.

Explicit vapor mass and this material-temperature bookkeeping extend the vapor-heat-only approximation discussed with equation 43 in Lin et al. (1983). They should be described as Nimbus's declared extension, not attributed as an unchanged published formula. The collection efficiencies and liquid shedding retain their stated published basis, while collision destination routing and number caps retain the approved Nimbus choices. Independent liquid-surface tests and frozen-admission tests check material paths, gross donors, moments, branch limits and energy accounting.

5.11 Share vapor before deciding the final thermal state

Several processes can consume the same vapor. Evaluating every process against the entire initial reservoir and adding the results can overdraw it even when each individual rate is donor-limited. The bounded joint owner therefore admits competing original-vapor demands together. Its current composition includes wet-hail exchange, zero-riming snow deposition and a later liquid-cloud equilibrium solve. It suppresses a simultaneous dry-hail diffusion source.

Surface flux rates use the entry state, and collected cloud comes only from the admitted original cloud reservoir. The cloud equilibrium problem is then solved at the source/EOS thermal endpoint. Newly created equilibrium cloud cannot retroactively supply captured collection. Likewise, vapor released by surface evaporation cannot be reused to increase same-interval snow or cloud condensation beyond their original-vapor bounds. After admission, the surface heat $L_v C$ is recomputed from the actual signed vapor amount.

The cloud solve has both equilibrium and reservoir constraints. If cloud remains, its vapor must match the represented liquid saturation condition. If all admissible cloud evaporates, a subsaturated endpoint can be valid. If the admissible original budgets cannot provide an acceptable equilibrium root, the owner rejects; it does not hide residual supersaturation with a clip. The equation uses the chosen energy endpoint and pressure, so it cannot be replaced by a saturation test at the original temperature.

The current accepted joint fixtures cover wet freezing and both signs of surface and cloud exchange. Shared-vapor saturation caps have helper-level coverage; that must not be advertised as an accepted cap-active full transaction. New joint melting/time-convergence, snow sublimation and remaining ice/graupel/general cloud collection or vapor owners are unfinished, along with live field integration. The joint source and joint tests are the durable description of this bounded scope.

5.12 The coupled source/EOS energy currency

Nimbus's basis-v2 thermodynamic inventory uses constant heat capacities and a common phase reference. Per kg dry air, write

$$\begin{aligned} u(T,q) = {} & c_{vd}(T - T_r) \\ & + [c_{vv}q_v + c_l(q_c + q_r) + c_i(q_i + q_s + q_g + q_h)](T - T_r) \\ & + (L_v - R_v T_r)q_v - L_f(q_i + q_s + q_g + q_h). \end{aligned}$$

The vapor reference differs from its latent enthalpy reference because internal energy excludes the gas pressure-volume contribution. For fixed composition, this inventory is affine in temperature; its slope is the mixture heat capacity per kg dry air. For fixed temperature, it is affine in the water amounts. Those two properties permit an explicit coupled endpoint without inserting an undocumented local c_p or c_v correction.

First compute the no-source EOS reference temperature T_{refEOS} from the original stored density and theta. This is distinct from diagnosing temperature using an original pressure that may be a stage snapshot. For an accepted composition q_1, the ideal closed cell-source endpoint is chosen to satisfy

$$\begin{aligned} u(T_*, q_1) &= u(T_{\text{refEOS}}, q_0), \\ T_* &= T_{\text{refEOS}} - \frac{u(T_{\text{refEOS}}, q_1) - u(T_{\text{refEOS}}, q_0)}{C_v(q_1)}. \end{aligned}$$

The owner then finds the theta whose subsequent production dry-EOS projection reaches this temperature at the held dry density. It stages source and EOS states separately, requires the expected sequence and rejects intervening mutations. Water storage rounding and theta/pressure storage rounding have separate receipt terms. The reference EOS change is retained as its own contribution rather than misidentified as a microphysical phase source.

This constructs an internal-energy mapping consistent with the declared model currency. Comparing the explicit material enthalpy path to that currency subtracts the change in $(R_d + R_v q_v)T$ per kg dry air, neglecting condensed-phase volume. This is an algebraic pressure-volume convention for the held dry-mass cell. It is not a validated full mechanical/thermal transport coupling for an open, compressible domain.

The source/EOS transaction is tested against the actual production EOS and independent energy evaluations in the permanent tests. The wet-surface tests additionally check a material-temperature path, because an accounting equality by itself cannot establish that an interfacial heat law is physically justified. The live solver has further operators between its microphysics source and final EOS projection. Owning their sequence and energy exchange remains required before this bounded mapping can become a complete runtime source owner.

Chapter 6

Radiation: from optical properties to a thermal source

Radiation changes the atmosphere by moving energy across its boundaries and redistributing absorption and emission within it. It does not transport dry-air mass. The selectable Nimbus families are `none`, `simple_grey` and `rrtmg`. The disabled family provides no radiative source. The other two make substantially different optical approximations, even though both ultimately provide temperature tendencies and flux diagnostics to the same runtime owner.

6.1 Define the sign, location and units before taking a divergence

Let $F^{\uparrow}$ and $F^{\downarrow}$ be nonnegative upward and downward flux magnitudes in $\mathrm{W\,m^{-2}}$, and define upward-positive net flux $F = F^{\uparrow} - F^{\downarrow}$. Fluxes live at layer interfaces. In a flat column with layer thickness h_k, the retained temperature conversion is

$$\dot{T}_k = -\frac{F_{k+1/2} - F_{k-1/2}}{\rho_{d,k} c_p h_k}.$$

Here the runtime supplies dry density, and c_p is the declared constant heat capacity used by this radiation interface. An increase of upward net flux through the layer means that energy leaves the layer, giving cooling. Reversing the flux sign without reversing the divergence is a physical sign error, not a choice of display convention.

Multiplying by dry layer mass per unit horizontal area gives

$$\rho_{d,k} c_p h_k \dot{T}_k = F_{k-1/2} - F_{k+1/2}.$$

Summing over layers cancels every interior interface. The atmospheric power per unit area is the bottom net flux minus the top net flux. This provides an independent interface-flux check on the column tendency. It does not prove the correctness of the optical coefficients or the chosen thermodynamic currency. The column evaluator checks this closure before publishing the float theta tendency and cell power.

The conversion supplied to the dynamics owner is $\dot{\theta} = \dot{T}(p_{\mathrm{ref}}/p)^{R_d/c_p}$ using pressure at the evaluated state. The later EOS projection still participates in the thermal path described in the coupling chapter. Column power, temperature tendency and the stored theta increment are related quantities, not interchangeable numerical arrays. For a terrain cell, physical heights, interface geometry, volume and surface area must retain their declared meaning before applying an analogous budget.

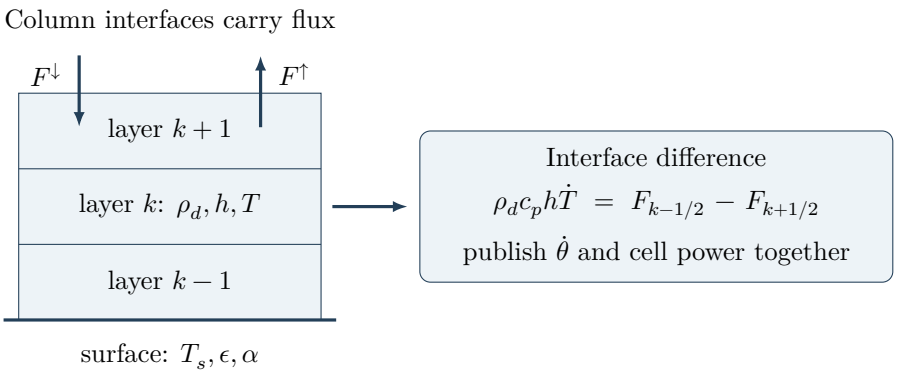


Figure 6.1: Upward-positive net flux is $F = F^{\uparrow} - F^{\downarrow}$. The layer gains energy when more net power enters its lower face than leaves its upper face. Summing layers cancels interior exchanges. Optical closure, surface boundary conditions and physical layer mass remain separate assumptions.

6.2 Optical depth summarizes an interaction along a path

At a frequency ν, let $I_\nu(x, \Omega)$ be radiance in direction Ω. A local, stationary transfer equation with elastic scattering can be written schematically as

$$\Omega \cdot \nabla I_\nu = -\beta_{e,\nu} I_\nu + \beta_{a,\nu} B_\nu(T) + \frac{\beta_{s,\nu}}{4\pi} \int_{4\pi} P_\nu(\Omega, \Omega') I_\nu(\Omega')\, d\Omega'.$$

Absorption and scattering coefficients have units m^{-1}; $\beta_e = \beta_a + \beta_s$. This convention normalizes the phase function so its angular integral is 4π. The thermal term assumes local thermodynamic equilibrium for the emitting material. A model that drops the scattering term has made a different physical approximation from one that replaces its angular integral with a quadrature.

For pure absorption along path coordinate s, optical depth is $\tau = \int \beta_a\, ds$. With no emission on that path, the solution is $I = I_0 e^{-\tau}$. A vertical optical depth $\Delta\tau$ seen by a direct beam with vertical cosine $\mu_0 > 0$ becomes path depth $\Delta\tau/\mu_0$ in a plane-parallel layer. This geometric factor belongs to the beam direction. A diffuse hemisphere contains many path angles and cannot generally use the same single cosine.

Nimbus's schemes reduce spectral and angular detail in different ways. The simple-grey scheme uses broadband optical depths. RRTMG uses spectral bands and reduced absorption-distribution quadrature. The underlying transfer equations help identify assumptions, but the code's quadrature, boundary rules and coefficient inputs define the numerical calculation actually performed.

6.3 The two simple-grey optical models

The pressure-only clear-sky model defines an interface optical-depth profile

$$\tau(p) = \tau_{\mathrm{ref}} \left(\frac{p}{100000\,\mathrm{Pa}} \right)^n, \qquad \Delta\tau_k = \tau(p_{k-1/2}) - \tau(p_{k+1/2}).$$

Longwave and shortwave have separate reference depths and exponents. Pressure interfaces must be finite, nonnegative and nonincreasing upward, with positive surface pressure. This is a prescribed broadband profile, not a calculation of molecular absorption from a complete gas spectrum. It deliberately provides a simple radiative experiment whose parameters are model assumptions.

The absorbing water/cloud-path model adds absorber mass paths,

$$\Delta\tau_k^{\text{water}} = \rho_{d,k} h_k (\kappa_v q_{v,k} + \kappa_l q_{c,k} + \kappa_i q_{i,k}).$$

The mass-absorption coefficients κ_x have units $\mathrm{m^2\,kg^{-1}}$, so the expression is dimensionless. Separate coefficients are supplied for the two broadband components. This path uses vapor, cloud liquid and cloud ice. It does not automatically turn rain, snow, graupel or hail into optical contributors merely because those categories exist in microphysics.

The absorbing model adds this term to the pressure-profile layer depth. It is not a scattering cloud model: changing cloud water changes absorption, but does not create the cloud scattering and angular redistribution represented by a more elaborate optical scheme. Configuration must pair the selected optical model with its clear-sky setting consistently. The simple-grey owner validates those choices and requires physical column inputs.

6.4 Grey longwave transfer and the surface boundary

The implemented longwave calculation treats each layer as isothermal for its broadband source. With $\mathcal{T}_k = e^{-\Delta\tau_k}$, it uses recurrences of the form

$$F^{\downarrow}_{k-1/2} = \mathcal{T}_k F^{\downarrow}_{k+1/2} + (1-\mathcal{T}_k)\sigma T_k^4,$$
$$F^{\uparrow}_{k+1/2} = \mathcal{T}_k F^{\uparrow}_{k-1/2} + (1-\mathcal{T}_k)\sigma T_k^4.$$

These equations describe the code's effective optical-depth convention; no additional longwave diffusivity factor should be inserted when reproducing this particular grey calculation. They assume a broadband thermal source and neglect longwave scattering in this path.

The top incoming longwave flux is zero. After the downward sweep reaches the surface, the upward lower boundary is

$$F^{\uparrow}_s = \epsilon\sigma T_s^4 + (1-\epsilon)F^{\downarrow}_s.$$

The second term represents reflected downwelling longwave. Omitting it would make a nonblack surface absorb energy that its declared emissivity does not permit under this opaque-surface convention. The upward sweep therefore depends on the completed downward sweep. Surface temperature and emissivity are explicit inputs; the radiation calculation does not solve a soil or land-surface energy balance that prognoses a new T_s.

The recurrences and the flux-divergence conversion are separate stages in radiative transfer. A valid recurrence can still produce the wrong heating if layer indexing, interface orientation or the mass used by the conversion is inconsistent.

6.5 Direct sunlight and Lambertian reflection are different angular fields

For an illuminated flat column, the top incident shortwave flux is $F^{\downarrow}_{\text{top}} = \mu_0 S_0$, where S_0 is irradiance normal to the solar beam. The downward grey path attenuates each layer by $e^{-\Delta\tau_k/\mu_0}$. The nighttime branch publishes no shortwave source rather than evaluating a singular or negative path length at $\mu_0 \le 0$.

At the surface, the reflected upward flux is $F_s^{\uparrow} = \alpha F_s^{\downarrow}$. Nimbus treats this reflected boundary as Lambertian. Its radiance is direction-independent over the upward hemisphere, so the upward flux at cumulative optical depth τ above the surface is

$$F^{\uparrow}(\tau) = F_s^{\uparrow}\, 2E_3(\tau), \qquad E_3(\tau) = \int_1^{\infty} \frac{e^{-\tau t}}{t^3}\, dt.$$

The factor follows directly from hemispheric integration: $2\int_0^1 \mu e^{-\tau/\mu}\, d\mu = 2E_3(\tau)$. It approaches one at zero optical depth. Applying the direct solar cosine to this reflected flux would give it the wrong angular distribution.

This grey construction contains atmospheric absorption and a reflecting lower boundary but no atmospheric scattering. Albedo changes the energy partition between the surface and the upward radiation; it is not a source of energy. The implementation evaluates the exponential-integral transmission with bounded numerical branches and rejects nonphysical results. Tests of the thin, thick, black-surface and reflecting-surface limits examine different parts of this contract.

6.6 What the correlated-k approximation changes

Molecular absorption varies rapidly with frequency. Resolving every spectral line in every grid column would be expensive. Within a band, an absorption coefficient distribution can be represented by a cumulative variable g, turning a band transmission integral into an approximation such as

$$\overline{\mathcal{T}}(u) = \int_0^1 e^{-k(g)u}\, dg \simeq \sum_j w_j e^{-k_j u}.$$

Here u is the absorber amount in the coefficient convention and the weights cover the distribution interval. Using correlated ordering between layers, combining gases and interpolating coefficients across temperature and pressure introduce additional modeling assumptions. This is a spectral quadrature; its g coordinate is not gravitational acceleration or a particle-size moment.

Mlawer et al. (1997) develops the longwave correlated-k reference method, and Iacono et al. (2008) describes the AER model family and its greenhouse-gas forcing calculations. Their validation results do not automatically apply to Nimbus's translated implementation, coefficient versions, cloud assumptions or coupled runtime.

Nimbus implements the calculation in C++; the AER netCDF files provide coefficients rather than an executable Fortran solver. Typed loading validates the expected coefficient schema and identity. Pressure/temperature preparation, major and minor gas absorption, continua, cloud optics and transfer have separate source owners under the RRTMG implementation. The reduced spectral structure is part of those coefficient contracts, not a tunable number of arbitrary equal-frequency samples.

6.7 Gas columns, cloud optical properties and transfer closure

Nimbus constructs dry-air molecular columns from the supplied dry density and physical layer thickness. With dry molar mass M_d in $\mathrm{kg\,mol^{-1}}$,

$$N_d = \frac{\rho_d h}{M_d}\,\frac{N_A}{10^4}$$

gives molecules per square centimetre. A water dry-air mass mixing ratio is converted to a dry-air-relative molar ratio with $r_v = q_v M_d / M_v$. Gas mole fractions and their allowed totals have explicit validation. Pressure, temperature and molecular column amounts therefore have different roles; substituting a pressure-thickness formula for the actual $\rho_d h$ preparation without reviewing its assumptions changes the inputs.

The runtime currently passes cloud liquid q_c and cloud ice q_i to the optics. In cloudy mode it uses liquid/ice mass paths $1000\rho_d h q_c$ and $1000\rho_d h q_i$ in $\mathrm{g\,m^{-2}}$, with configured effective radii in micrometres. A cell with either positive cloud category is assigned cloud fraction one; otherwise it is clear. There is no prognostic subgrid cloud fraction or general cloud-overlap model in this construction. Effective radii are optical inputs here, not automatically the radius diagnosed from every microphysics moment. Falling condensate categories are not silently summed into the two cloud paths.

Longwave transfer includes its band-dependent diffusivity treatment. Shortwave transfer uses the implemented practical improved flux method (PIFM) two-stream layer coefficients, direct/diffuse reflection and transmission, and the supplied scattering/asymmetry information. A two-stream solution represents an angular closure, not the full radiance field at every direction. Conservative scattering and optically thin limits need stable numerical branches because algebraically equivalent expressions can suffer cancellation or singular-looking intermediate terms. See longwave transfer and shortwave transfer.

Interface temperatures, when not supplied, use the retained pressure-coordinate interpolation and endpoint rules. The physical column order and the transfer solver's sweep order are explicitly converted. Agreement of a single surface flux does not establish that all interface fluxes, layer heating and cloud/gas responses agree with an independent column reference.

6.8 Terrain adds visibility, projected area and horizontal exchange

For a unit direction toward the Sun, s, and an upward surface normal n, direct intercepted irradiance is proportional to $\max(0, n \cdot s)$. Visibility is an additional condition: a face aimed toward the Sun may still be shadowed by intervening terrain. Area measured along the slope differs from horizontal plan area. Fluxes in $\mathrm{W\,m^{-2}}$ must be paired with the area their definition uses before becoming power in watts.

Nimbus has explicit terrain shortwave paths for visibility, ray traversal, absorbed power, reflected angular transport and spectral variants. A ray through several physical cells carries an energy amount; attenuation along each segment partitions it between absorption and outgoing power. A reflecting terrain boundary can redirect it into other columns. Reusing independent vertical-column heating while merely shading a picture would not represent that exchange.

For a bounded shortwave transport calculation, an appropriate whole-domain partition is

$$P_{\mathrm{in}} = P_{\mathrm{atmosphere}} + P_{\mathrm{surface}} + P_{\mathrm{top\ escape}} + P_{\mathrm{lateral\ escape}} + P_{\mathrm{unresolved}}.$$

The unresolved term records power remaining when a declared angular/bounce or convergence bound is reached; it is not automatically atmospheric heating. Changing quadrature, reflection count or stopping tolerance changes a numerical approximation that must be checked independently of this bookkeeping identity.

Source presence does not imply every terrain/backend combination is admitted. The current runtime rejects mapped RRTMG shortwave on an active accelerator because its native spectral-terrain

owner is missing. CPU mapped spectral work and native column batches have separate scopes. The radiation runtime is the authority for the actual selected path and required rejection behavior.

6.9 Cadence, stored power and the complete coupled step

Radiation is refreshed on its declared cadence and can hold its source between calls. The evaluated optical state, published heating and held power must share one completed generation. The next dynamics interval should not combine a newly computed power diagnostic with an older held theta tendency. A refresh failure must preserve the preceding coherent runtime state and its timing.

If a held atmospheric power is P_h over an interval Δt, its declared energy input is $P_h \Delta t$. The radiation energy ledger accounts for that held input and the relevant source endpoint. This is distinct from simply recomputing an interface-flux difference from a later atmospheric column. Cadence holding approximates time evolution; a smaller dynamics step does not by itself make a long-held radiation source current.

The checks in radiation tests and runtime-state tests cover columns, units, invalid inputs, closure and publication behavior for their stated scopes. Broader coupled, terrain, restart and native paths need their own bounded and campaign evidence. Neither a familiar scheme name nor a finite flux residual replaces that validation.

Chapter 7

Unresolved mixing, stochastic forcing and diagnostic interpretation

The resolved equations do not describe every turbulent eddy, surface exchange or particle-scattering process inside a cell. Closures supply those missing relationships. They are mathematical models with their own state, scales and admissible inputs, not interchangeable corrections to a finished solution. This chapter follows the implemented Nimbus closures and their connections to transport and diagnostics. The earlier grid and microphysics chapters supply the carrier and moment conventions; the architecture guide explains the corresponding state and execution owners.

7.1 Eddy viscosity connects resolved strain to unresolved transport

For physical Cartesian velocity components, define the symmetric strain $S_{ij} = \frac{1}{2}(\partial_j u_i + \partial_i u_j)$ and its magnitude $|S| = (2S_{ij}S_{ij})^{1/2}$. Strain has units s^{-1}. An eddy viscosity models an unresolved momentum exchange through a stress proportional to resolved deformation. A viscous-traction form is $\tau_{ij} = 2\rho_d \nu_t S_{ij}$, with acceleration from its divergence divided by the appropriate carrier. This motivates the closure; its isotropic/compressional and directional treatment must still follow the actual discrete tensor owner.

Static Smagorinsky uses

$$\nu_t = (C_s\Delta)^2|S|, \qquad K_\theta = \nu_t/\mathrm{Pr}_t, \qquad K_q = \nu_t/\mathrm{Sc}_t.$$

Here C_s is dimensionless, Δ is a length, and viscosity/diffusivities are in $\mathrm{m}^2\ \mathrm{s}^{-1}$. Nimbus additionally applies configured coefficient bounds and the selected stability correction. Increasing C_s or the filter length increases modeled mixing; it is not equivalent to changing a plotting resolution. The retained `dx` filter option actually uses $\sqrt{\Delta x \Delta y}$, while `cubic_root` uses $(\Delta x \Delta y \Delta z)^{1/3}$. A user filter is a prescribed physical length. Mapped filters use the horizontal footprint and physical Jacobian thickness, not a sloping face-normal CFL distance.

Directional modes select which gradients and exchanges participate. In the flat derivative limit, the strain measures have the forms

$$|S|_h^2 = 2(S_{xx}^2 + S_{yy}^2) + 4S_{xy}^2 + (\partial_x w)^2 + (\partial_y w)^2,$$
$$|S|_v^2 = (\partial_z u)^2 + (\partial_z v)^2 + 2S_{zz}^2.$$

The horizontal-only mode therefore retains horizontal gradients of vertical velocity. It is not simply a two-component two-dimensional atmosphere. The C-grid implementation averages appropriate edge contributions; averaging a gradient before squaring is generally different from averaging its square. These choices are explicit in the shared SGS operators.

For example, two neighboring shear samples $+a$ and $-a$ have a zero mean but a mean square a^2. A closure that first averages them and then squares would diagnose no deformation contribution at all. This is why matching array sizes does not justify changing the staggered averaging order. The physical interpretation is also limited: eddy viscosity parameterizes unresolved transport at a filter scale; it is not molecular viscosity and does not resolve the omitted eddies. Refining the grid while retaining a grid-derived Δ changes both the resolved field and its unresolved closure.

Stable stratification can suppress turbulence. With the declared buoyancy frequency squared N^2, the versioned Lilly-equilibrium branch uses a strain factor equivalent to

$$|S|_{\text{eff}} = \sqrt{\max(0, |S|^2 - N^2/\text{Pr}_t)}.$$

Positive N^2 reduces mixing; negative N^2 increases the available scale. Nimbus evaluates the expression with separate stable/unstable numerical forms and retains a distinct older Richardson-correction option. Those options must not be called identical. Moist versus dry buoyancy, coefficient caps and the selected directional mode are part of the closure contract. The static vertical-only tracked scope remains Engineering incomplete; the presence of these formulas does not authorize every PBL, terrain or accelerator combination.

7.2 Dynamic Smagorinsky diagnoses a bounded coefficient

A dynamic model uses information at a second filter scale rather than prescribing one coefficient everywhere. Nimbus reconstructs scalar-center fields, applies a density-weighted Favre test filter and forms a resolved stress difference L_{ij} and a model difference M_{ij}. With a local least-squares assumption, the coefficient multiplying the grid-scale model is estimated from their contraction. The retained implementation averages horizontally at each height:

$$C(k) = \text{clip}\left(\frac{\sum_{i,j} L_{ab} M_{ab}}{\sum_{i,j} M_{ab} M_{ab}}, 0, C_{s,\max}^2\right), \qquad C_s(k) = \sqrt{C(k)}.$$

Repeated tensor indices denote the symmetric-tensor contraction, including off-diagonal contributions. If the denominator is too small, the coefficient is zero. The nonnegative bound excludes SGS backscatter in this implementation; it is a deliberate model restriction, not a general theorem about turbulence.

The test width is twice the grid filter width. Favre filtering uses $\widehat{\rho_d \phi}/\widehat{\rho_d}$, so ordinary averaging is not an equivalent replacement in a variable-density field. Horizontal periodic wrapping and bounded vertical sampling are also part of the retained filter. The current owner requires its flat, periodic, C-grid, 3-D CPU domain and rejects a required-native request. It is not a generic dynamic model for all boundaries or mapped geometries. See dynamic coefficient construction and SGS reference tests.

Horizontal averaging is a substantive assumption. It supplies one coefficient for a height plane and suppresses fluctuations in the least-squares ratio, but can blend a strongly sheared storm core with a quieter environment. A pointwise or Lagrangian average would define a different model, not simply a faster reduction. The current test filter has separable three-point weights $(1/4, 1/2, 1/4)$;

its finite vertical edge treatment is distinct from a periodic three-dimensional filter. The dynamic branch also rejects a separately selected stability correction rather than composing two coefficient models without a declared derivation.

7.3 Prognostic TKE supplies a second modeled energy reservoir

Let e denote unresolved turbulent kinetic energy per unit dry mass, with units $\mathrm{m^2\ s^{-2}}$. Its closure has the schematic material form

$$\frac{De}{Dt} = P_s + P_b - \epsilon + \frac{1}{\rho_d}\nabla\cdot(\rho_d K_e \nabla e).$$

Shear production transfers energy through deformation; buoyancy production can create or suppress turbulence; dissipation removes modeled TKE. Resolved advection and source application still belong to the coupled timestep. A TKE equation alone does not establish a conservative exchange with resolved kinetic or internal energy.

Nimbus's Deardorff–Moeng-style branch uses $\nu_t = 0.1\ell\sqrt{e}$. With its stability correction active, the stable mixing length is limited by $\ell = \min(\Delta, 0.76\sqrt{e}/N)$ for $N > 0$; neutral/unstable conditions use the grid length, and disabling the correction retains that length directly. For positive e, its retained dissipation is

$$\epsilon = \left(0.19 + 0.51\frac{\ell}{\Delta}\right)\frac{e^{3/2}}{\ell}.$$

The zero-energy case has its own zero-source treatment. This avoids interpreting the displayed quotient as permission to divide by a vanished length.

Scalar transport also depends on the length ratio. Before configured caps,

$$K_\theta = \nu_t\left[1 + (\mathrm{Pr}_t^{-1} - 1)\ell/\Delta\right],$$
$$K_q = \nu_t\left[1 + (\mathrm{Sc}_t^{-1} - 1)\ell/\Delta\right], \qquad K_e = 2\nu_t.$$

Thus replacing all three by one diffusivity changes the closure. Stress work, TKE production and scalar diffusion use physical geometry and the admitted discrete locations. The TKE source and tests define the actual branch, including finite input, positivity and native-path requirements.

The stable length has the expected dimensions: $\sqrt{e}$ is a velocity and N an inverse time, so their ratio is a length. As stability reduces ℓ/Δ, the scalar-to-momentum ratios above approach one. This is different from applying a constant Prandtl number after computing viscosity. The dissipation coefficient also varies with the length ratio, so changing only the viscosity length would leave an inconsistent closure. Whether the removed TKE is returned to another represented energy reservoir must be checked in the actual coupled source/energy owner; the sign of ϵ alone does not prove that exchange is implemented conservatively.

7.4 PBL columns and surface exchange operate at another modeling scale

The PBL choices are none, slab, YSU and MYNN. None supplies no PBL forcing; it does not disable separately selected resolved-grid SGS. Active PBL models represent vertical column transport and surface exchange with different assumptions from a three-dimensional LES stress. Combining them requires an explicit division of transport to avoid counting the same unresolved exchange twice.

A useful column notation makes the nonlocal assumption visible:

$$J_\phi = -\rho_d K_\phi(\partial_z \phi - \gamma_\phi) + J_{\phi,\text{entrainment}}.$$

γ_ϕ has the units of a vertical gradient. For theta, J_θ has units kg K m^{-2} s^{-1} and still needs the declared heat conversion to become W m^{-2}. For a water mixing ratio, the analogous flux is a water mass flux. A positive countergradient contribution can transport a scalar against its local gradient because it represents nonlocal eddies. Consequently, one cannot infer the complete flux sign from $K > 0$ alone. This schematic form organizes the coupling; each PBL family supplies its own terms and limits.

The slab owner retains a mixed-layer state and diagnoses a bounded depth h. Its column heights are measured above local ground. Weighting respects partial layer overlap:

$$m_k = \rho_{d,k} \max\Big(0, \min(h, z_{k+1/2}) - z_{k-1/2}\Big), \qquad \overline{\phi}_m = \frac{\sum_k m_k \phi_k}{\sum_k m_k}.$$

These are masses per horizontal area. Mixing and surface forcing are applied using complete cell masses so a partly occupied top layer does not receive a full layer's forcing. Stored mixed-layer quantities and realized fluxes must remain consistent with the resolved column. The word slab here does not imply a prognostic soil reservoir. See slab mixing.

YSU supplies nonlocal transport and explicit entrainment, with physical background in Hong, Noh and Dudhia (2006). Nimbus's interior diffusivity profile has the characteristic $K \sim \kappa w_m z(1-z/h)^2$ structure before its stability and bounding choices. A scalar flux contains a downgradient part and, where selected, a countergradient correction; entrainment adds a distinct top exchange. The implemented temperature, moisture and momentum terms are not all identical: for example, its moisture flux does not simply reuse the temperature countergradient term. The retained YSU owner determines the exact coefficients and interval solve.

MYNN uses prognostic TKE and stability functions, with background in Nakanishi and Niino (2009). In Nimbus's level-2.5 notation, $q^2 = 2e$ is a velocity-variance quantity, not water mixing ratio. The diffusivities have the form

$$K_m = \ell\sqrt{2e}\, S_m, \qquad K_h = \ell\sqrt{2e}\, S_h,$$

with stability functions depending on shear, buoyancy and the mixing-length/TKE state. The retained Helfand–Labraga correction limits unrealizable states; invalid denominators or coefficients reject rather than silently producing an arbitrary mixing rate. MYNN's column TKE is a closure state with its own solve, not permission to add an independent LES TKE source twice. See MYNN and stability functions.

7.5 Surface-flux signs, units and prescribed skin state

Let U be near-surface air velocity relative to the physical surface, $U = |U|$, and $\Pi = (p/p_{\text{ref}})^{R_d/c_{pd}}$. The retained bulk exchange takes the form

$$\begin{aligned} \tau &= -\rho_d C_d U U, \\ H &= \rho_d c_{pd} \Pi C_h U(\theta_s - \theta_a), \\ E &= \rho_d C_e U(q_s - q_a). \end{aligned}$$

Stress opposes relative air motion; positive H heats the atmosphere and positive E supplies vapor. Their units are N m^{-2}, W m^{-2} and kg m^{-2} s^{-1} respectively. $L_v E$ is a latent-energy flux, whereas E itself is not. The Exner factor in H converts a potential-temperature flux to sensible-heat currency; omitting it changes the implemented exchange.

Neutral coefficients use separate momentum and heat roughnesses:

$$C_{d0} = \frac{\kappa^2}{\ln^2(z/z_{0m})}, \qquad C_{h0} = \frac{\kappa^2}{\ln(z/z_{0m})\ln(z/z_{0h})}.$$

The bulk branch applies retained Louis stability factors. The Monin–Obukhov branch instead iterates stability-corrected logarithmic denominators using $\zeta = z/L$ and the surface virtual-potential-temperature flux:

$$L = -\frac{u_*^3 \theta_{v,a}}{\kappa g \, \overline{w'\theta'_v}_s}, \qquad \overline{w'\theta'_v}_s \simeq \frac{H}{\rho_d c_{pd} \Pi} + 0.61 \theta_a \frac{E}{\rho_d}.$$

An unstable positive virtual heat flux therefore gives negative L. Nimbus retains explicit neutral limits, iteration limits and minimum-friction-velocity handling; these are part of the algorithm, not an unbounded analytic law.

The minimum-u_* treatment can increase momentum and heat/moisture transfer coefficients together. It is therefore a modification of the exchange, not a harmless display floor. Roughness must be below the reference sampling height, and the neutral limit must remain finite. These checks are necessary even for a flat surface: a logarithmic formula evaluated outside its geometric domain does not become physical by clipping its final flux afterward.

Skin temperature and moisture are prescribed, or initialized from the initial surface air when unset and subsequently held. There is no prognostic soil heat/moisture or vegetation model behind them. On sloping ground, tangent wind, traction and actual-versus-horizontal area must be transformed consistently. See surface exchange. Neither radiative surface absorption nor a surface-flux receipt automatically evolves a land temperature that the model does not own.

7.6 Diffusion applies a closure; it does not define one

The density-weighted scalar diffusion form is $\rho_d \partial_t \phi = \nabla\cdot(\rho_d K \nabla\phi)$. The coefficient can come from a prescribed numerical profile or a physical closure. Explicit time integration has a diffusion stability restriction; implicit vertical exchange solves a coupled column system over the interval. Neither method makes an inappropriate coefficient physically justified.

Nimbus separates coefficient construction, transport/application and realized source reports. An implicit solve's accepted flux need not equal an explicit flux formed from the old gradient. PBL held sources retain interval-mean forces and inventory rates so changing density does not accidentally reuse a stale acceleration denominator. Momentum diffusion also changes resolved kinetic energy, with a finite-step contribution beyond linear work. Appropriate boundary work and TKE/thermal attribution must be assessed together. The existing diffusion chapter derives the spatial and temporal operators; the closure-specific question is which K, carrier, interval and coupled energy owner feed them.

Some admitted paths defer transport-tendency construction until the selected application owner consumes closure coefficients. That prevents an explicit tendency from being applied and then implicitly applied again over the same interval. It also makes coefficient lifetime significant: a matrix built from one state is not automatically the matrix corresponding to a later gradient or

density. Changes between explicit, implicit and deferred paths need tests of the accepted state and realized flux, including rejection, rather than a comparison of coefficient arrays alone.

7.7 Stochastic perturbations have covariance and conservation contracts

The chaos choices are none, initial-condition perturbations, boundary-layer perturbations and full stochastic coupling. A simple temporal building block is the retained AR(1) update

$$\xi_{n+1} = a\xi_n + \sqrt{1-a^2}\,\eta_n, \qquad a = \exp(-\Delta t/\tau).$$

If independent innovations η_n and the previous field each have zero mean and unit variance, the new variance is $a^2 + (1-a^2) = 1$. That explains the innovation coefficient. Finite-grid normalization, spatial filtering, clipping and conditioning can change the realized distribution. The variance identity does not establish the exact finite-domain distribution after those additional transformations. The actual noise refresh interval must be used; replacing it with an unrelated solver step changes decorrelation.

The spectral spatial filter multiplies Fourier components by

$$G(k_x, k_y) = \exp[-\tfrac{1}{2}\{(L_x k_x)^2 + (L_y k_y)^2\}].$$

This attenuates short scales. The recursive alternative applies repeated forward/backward smoothing with weights such as $\exp(-\Delta x/L_x)$. It approximates a correlated field through a different discrete operation; it is not identical to the spectral filter. Filter widths, boundary treatment and variance normalization affect the resulting covariance. A configured length is not automatically the measured autocorrelation length of every clipped or relocated field. See correlation filters and temporal updates.

Seeds are combined with member, process/field and sequence identity. Restart must preserve the retained noise generation and clock. Earth-correlated noise uses its declared physical control-grid coordinates; a translated window must not invent a new forcing realization solely because an Earth point acquires a different array index. Different standard-library random-distribution implementations are not automatically bitwise equivalent despite equal seeds.

A source multiplier of the form $1 + \alpha\xi$, subject to the declared bounds, changes a process. It must be applied to the associated transfers or conserved column exchange consistently. Scaling water tendencies independently can break their zero sum; scaling a sedimentation divergence without its lower-boundary flux breaks precipitation accounting. Nimbus retains source/column ownership and private-state checks for admitted stochastic couplings. The current full-stochastic owner is not a license to multiply every physical term by unrelated noise. Ensemble spread is also not calibrated forecast uncertainty merely because the perturbation variance is known.

Bounding a multiplier also changes its statistics: even if ξ has zero mean, the mean of a clipped $1 + \alpha\xi$ need not be exactly one. Common process identity and a conserved-transfer construction address a different question from unbiased ensemble forcing; neither follows automatically from the AR(1) variance calculation.

7.8 Radar products are bulk forward operators

Radar diagnostics translate modeled particles and wind into an observation-like quantity. A Rayleigh liquid reference motivates the sixth-diameter moment $Z = \int D^6 n_v(D)\, dD$, with a vol-

ume spectrum and conversion to $\mathrm{mm}^6\ \mathrm{m}^{-3}$. Its logarithmic form is

$$Z_{\mathrm{dBZ}} = 10 \log_{10}(Z_e/(1\ \mathrm{mm}^6\,\mathrm{m}^{-3}))\,.$$

Frozen-particle density, dielectric behavior and shape require further assumptions. Summing component reflectivities must occur in linear space, not by adding their dBZ values.

Nimbus's `fast_da` and `psd_moment` tiers retain concrete bulk relationships. Without supplied number, a species path uses $Z_{e,x} = A_x(\rho_d q_x)^{1.75}$. With supplied number, its moment proxy has the form $Z_{e,x} = C_x(\rho_d q_x)^2/N_{v,x}$ with retained species factors. The number used by that interface is interpreted per volume; a private number per kg dry air needs conversion before being supplied. The tier can use the mass-only relation when the corresponding number is absent. These are retained proxies, not an arbitrary gamma-spectrum integral or a general electromagnetic scattering solver. See reflectivity tiers.

Radial velocity projects reconstructed physical wind onto the vector from the radar to the sample: $V_r = u \cdot \hat{r}$, with the configured bulk scatterer fall correction when selected. C-grid face winds are reconstructed at the declared scalar sample before projection. Terrain heights and Earth origin determine the line of sight. The current `beam_volume` option uses a bounded local neighborhood average; it is not a complete antenna beam, pulse-volume, attenuation or instrument-noise simulator. The velocity owner defines its sampling and fall-speed bounds.

Differential reflectivity is $Z_{DR} = 10 \log_{10}(Z_H/Z_V)$ where the signals are valid. Nimbus retains `simple`, `polarimetric_fo` and `zhang2021_rain_mp` paths. The bulk polarimetric path uses prescribed shape/species relationships and publishes associated KDP and correlation products within declared ranges; its name does not establish a T-matrix calculation. The separate rain-MP path rejects frozen-species participation. KDP is expressed in degrees per kilometre and the correlation coefficient is dimensionless; these units do not make a bulk proxy a resolved beam-phase model. Low/no-signal conventions and clipping are part of output interpretation. The polarimetric source and its tests must accompany any claim about an admitted product set.

7.9 Diagnostic integrals, storm objects and parcel paths

Budget diagnostics integrate physical inventories and boundary exchanges, using their versioned dry/mixture mass and phase-reference conventions. An exact telescoping ledger records how a diagnostic changed; independent source, work and boundary references are still needed to validate its physics. Derived gradients and layer integrals require physical heights and finite coverage, not a nominal index range.

The sounding diagnostic lifts a surface parcel dry-adiabatically to its LCL and with a retained moist lapse-rate approximation above it. Vapor-corrected virtual temperatures give buoyancy $B = g(T_{v,p} - T_{v,e})/T_{v,e}$. Its CAPE is the positive buoyancy integral between diagnosed LFC and EL; CIN is reported as a nonnegative magnitude of the negative integral below LFC. Nimbus uses discrete layer integration and finite column coverage. With no LFC it reports unavailable crossing heights and a no-LFC interpretation with zero CIN, not evidence that inhibition is physically absent. A top-limited EL likewise needs its finite-column context. See sounding diagnostics.

Storm detection uses resolved updraft, condensate and rotation evidence to form feature candidates. Temporal association, branching and ambiguity rules select a track rather than identifying a uniquely defined physical object from one maximum. Updraft helicity has the dimensional form $\int w\zeta\, dz$ over its declared updraft/height support; storm-relative helicity uses the signed storm-relative hodograph convention. Their values depend on physical layer coverage and the selected

storm motion. The tracker retains history because continuity cannot be reconstructed from a single frame.

Air-parcel trajectories solve $dx/dt = u(x, t)$ under the model's interpolation and frame assumptions. Nimbus prepares a predicted midpoint from the beginning wind, samples there, and combines that midpoint sample with the end-state midpoint wind during completion. This differs from a streamline in one frozen velocity field. Stagger-aware sampling, physical height, vertical clamping, horizontal departure and Earth-frame output are explicit choices; the paths are not inertial precipitation trajectories. Trajectory continuation retains parcels and sample cadence through restart and movement.

These diagnostics test different aspects of a simulation. Matching rainfall, CAPE or a storm track does not uniquely validate all closures that produced them. Bounded formula, coupling and failure checks establish their stated implementation scope; long campaigns and independent scientific review remain separate from those checks.

Chapter 8

Putting it all together: one atmosphere, one timestep

The preceding chapters describe individual equations and numerical choices. A simulation becomes coherent only when those choices describe the same air, water, geometry and time interval. This chapter follows a complete update and shows how to reason about its coupled behavior. The main example is the Cartesian C-grid split-explicit path with bulk microphysics. Other admitted integrators and geometries have their own stage owners; the example is not a claim that one ordering can be substituted for every supported configuration.

8.1 Begin with a complete mathematical state

A useful abstract state is

$$X = (\rho_d, \theta, u, q, N, Z, s, \mathcal{G}, \mathcal{H}).$$

Here q contains the selected water species, N contains the applicable number concentrations, and Z denotes additional prognostic moments only where a scheme actually owns them. s includes scheme-private physical state such as held column sources or turbulence variables. $\mathcal{G}$ is geometry and domain placement; $\mathcal{H}$ includes histories needed by boundaries, staged integration and continuation. Some of these components are absent for a simpler configuration. Pressure is additionally stored and used, but its committed relationship to dry density and potential temperature is constrained by the production EOS.

The state is larger than the arrays shown in a plot. Two runs with identical visible temperature and water at one instant can evolve differently if their particle number, held heating interval, radiative boundary history or nested placement differs. A restart specification must therefore identify the full continuation state. A snapshot of plotted fields is insufficient evidence that the next tendency will match.

Geometry is included because changing cell volume changes the interpretation of density and mixing ratio. Time history is included because a source held between radiation calls is not equivalent to a source freshly evaluated from the current cloud. A complete state definition is the first defense against accidentally comparing different mathematical problems while believing that only a backend, thread count or file format changed.

8.2 Initialization must create a mutually compatible state

A sounding supplies a vertical environment, but the solver must place it on its own scalar and velocity locations. The procedure needs physical heights, a pressure/base-state construction, water conventions and an equation of state. An interpolation that is monotone in a sounding coordinate does not necessarily preserve hydrostatic balance on the model interfaces. Similarly, using moisture in a thermodynamic conversion does not justify adding it a second time to a buoyancy correction.

For a simple resting dry column, hydrostatic balance is $dp_0/dz = -\rho_0 g$. A discrete balance might be expressed at a face as

$$\frac{p_{0,k+1} - p_{0,k}}{z_{k+1} - z_k} = -g\rho_{0,k+1/2},$$

provided this is the same averaging and distance used by the force operator. That last condition is essential. Integrating a continuous hydrostatic profile very accurately and sampling it with an incompatible force stencil can still launch adjustment waves. Nimbus's initialization and hydrostatic tests examine the implemented pairing; they are more informative than assuming that the word "hydrostatic" in a configuration guarantees a discrete resting solution.

A thermal perturbation is also a declared intervention in the initial state. Holding pressure fixed while increasing theta changes diagnosed temperature and density differently from holding density fixed. A bubble specification must be interpreted through the actual initialization path, not through an informal statement that it "adds heat." The initialization implementation, initialization tests and the public configuration compiler make those choices concrete. Invalid dimensions, missing coefficient tables, incompatible boundaries and unavailable combinations must be rejected before advancing a partially constructed state.

8.3 Choose the interval before assigning its sources

The outer runtime chooses an allowed large time step from configuration, current state and the relevant stability constraints. The acoustic substep count then belongs to the chosen split method. Output times, source cadences and a requested end time may further bound an interval. A process that receives a smaller interval must use that interval consistently for its tendency integration, precipitation and diagnostic receipts.

For a source evaluated every τ seconds, a held tendency can be written schematically as

$$S(t) = S(X(t_m)), \qquad t_m \le t < t_m + \tau.$$

This is an approximation to the continuously reevaluated source. Its error depends on how rapidly the source and its inputs vary during the hold, not merely on whether τ divides a nominal large step. For example, rapidly changing condensate can change radiative heating while the held value remains unchanged. A continuation must preserve both the held value and its timing. Reevaluating it immediately after restart changes the experiment.

Nested levels add a synchronization problem: a child's shorter steps cover a parent interval, but a parent source or boundary history is not automatically valid at each child stage. The execution plan must specify interpolation, refresh and feedback times. The public admission system exists partly because individual schemes being selectable does not establish those combined owners.

8.4 Transport dry air and its carried quantities coherently

For the representative split path, slow tendencies are prepared from the appropriate stage state. The integrator applies the slow increment, seeds the acoustic radiation history, and advances the fast pressure/density and momentum sequence through its acoustic substeps. Closure of the fields read by the next substep is part of that sequence. Reusing a host or device buffer is correct only if it holds the state required at this particular stage.

The continuity calculation produces a carrier mass change. Retained face fluxes allow the subsequent scalar update to use the same mass that moved through each face. For a species x, the coupled relation is

$$M_C^* = M_C^n - \sum_f B_{Cf} \mathcal{F}_f,$$

$$W_{x,C}^* = M_C^n q_{x,C}^n - \sum_f B_{Cf} \mathcal{F}_f q_{x,f},$$

$$q_{x,C}^* = W_{x,C}^* / M_C^*,$$

where $\mathcal{F}_f$ is the time-integrated dry-mass flow, in kg, over the interval used by this transport operation. The asterisk denotes the transported endpoint, not the final source-adjusted state. A retained flux integrated over one interval cannot be reused as though it represented a different interval without an explicit conversion justified by the integrator.

Private moments need the same care as visible water. If a number is stored per kg dry air, its inventory follows the dry-mass carrier. If it is stored per volume, its numerical transport must use that definition. Moving mass with one carrier while moving particle number with another can create a false change in mean size even if both arrays remain positive. This affects terminal velocity, collection and radiative interpretation at the next process call.

A basic invariant is the uniform-tracer test: a constant mixing ratio should remain constant when only its carrier moves. Another is a closed or periodic inventory test. A third is the boundary flux balance for an open domain. These tests answer different questions and should not be replaced by a visual check that a cloud retains roughly the same shape.

8.5 A microphysics call is a coupled material transaction

At a transported cell, microphysics receives pressure, temperature/theta, dry density and the selected water and moment state. Processes redistribute water among vapor, liquid and ice, change particle counts and sizes, exchange latent energy, and move precipitation vertically. A process list is not a set of independent increments that may all consume the original reservoir. Ordering, shared-vapor allocation, collection routing and moment repair define the actual closure.

Write the local phase-source inventory change as

$$\Delta W_x = M_C \Delta q_x.$$

With no external water source and no sedimentation across the cell boundary, the phase-change identity is $\sum_x \Delta W_x = 0$. A domain water budget must add sedimentation and other boundary exchanges. The appropriate total is all prognostic water categories, not just cloud condensate or precipitation. A cloud-to-rain conversion can reduce displayed cloud while conserving water. An evaporating raindrop can disappear from the rain field while increasing vapor and cooling the cell.

The thermodynamic increment follows the declared phase energies and heat capacities. For an illustrative constant-pressure condensation of a small amount δq with constant latent heat and heat capacity,

$$\Delta T \simeq \frac{L_v \delta q}{c_p}.$$

This is an explanatory approximation with held coefficients and the stated thermodynamic constraint. The earlier microphysics chapter describes more complete mixture and material-temperature issues. Freezing, melting, liquid surface temperature, vapor competition and EOS projection prevent this one-line estimate from being a universal energy owner.

A collection rule also makes a distributional assumption. Combining two mass reservoirs without a compatible number change alters the inferred spectrum. A count cap is therefore a model choice with consequences for size, fall speed and subsequent collection, not merely protection against a large integer. The documented Nimbus routing choices, including its retained new-hail gate and count bounds, must be stated when comparing the model with a publication. The guarded third-moment owners do not yet establish a complete live source/EOS transaction; their source presence is not public readiness.

8.6 Pressure adjustment is part of the thermal path

The production dry rho–theta EOS can be expressed, with $\kappa = R_d/c_p$ and $\gamma = 1/(1-\kappa)$, as

$$p = p_{\mathrm{ref}} \left(\frac{\rho_d R_d \theta}{p_{\mathrm{ref}}} \right)^{\gamma}, \qquad T = \theta \left(\frac{p}{p_{\mathrm{ref}}} \right)^{\kappa}.$$

These equations describe the declared dry pressure law. Moisture affects other parts of the model, including phase energy and loading; that does not make the pressure relation a full moist-mixture EOS.

The thermal sensitivity depends on what is held fixed. If pressure is held, $dT/T = d\theta/\theta$. If dry density is held and pressure is subsequently projected by the EOS, then

$$\frac{dT}{T} = \gamma \frac{d\theta}{\theta}.$$

Thus a theta increment designed using a fixed-pressure temperature conversion and a later fixed-density pressure projection form a different endpoint map from the original fixed-pressure calculation. The equation is a local sensitivity demonstration, not a claim that every Nimbus process applies that simplified source or has an error of exactly γ.

To prescribe a particular endpoint temperature at held dry density, one can instead solve the EOS relation for the required theta,

$$\theta = T^{1-\kappa} \left(\frac{p_{\mathrm{ref}}}{\rho_d R_d} \right)^{\kappa}.$$

But choosing that endpoint requires an appropriate energy constraint first. Replacing one conversion with this expression without deciding whether the physical process is a constant-volume, constant-pressure or open-system exchange would simply conceal a different assumption in the code.

Nimbus has explicit bounded work on a source/EOS energy mapping, described in the preceding chapter. The live orchestration can place SGS, diffusion and other operations between a microphysics source and final pressure projection. Their changes must be represented in the complete

endpoint transaction. A cell-level mapping that rejects intervening mutation cannot be treated as a complete field-level integration merely because its isolated residual is small.

8.7 Mixing, sedimentation and radiation close different budgets

Subgrid mixing exchanges momentum and scalar inventories between locations. The eddy diffusivity depends on the closure and sometimes on gradients of the state being mixed. A diffusivity has units of area per time; it is not itself a mass flux. Density, face area, gradient and the integration interval turn it into an exchange. A dissipative velocity update also needs a declared treatment of its kinetic-energy change. An energetically complete closure cannot be inferred solely from the sign of a diffusion coefficient.

Sedimentation is transport relative to the air. Its vertical flux depends on the category's mass or moment distribution and its terminal-velocity closure. The bottom boundary contributes precipitation leaving the atmosphere. If a surface accumulation is updated without removing the corresponding atmospheric inventory, water is created in the combined atmosphere-plus-surface accounting. If removal occurs without the accumulation receipt, the atmospheric budget may be correct while the published precipitation diagnostic is wrong.

For radiation, column heating follows flux divergence with the actual physical layer mass and the chosen heat-capacity convention. The surface and top-of-column radiative fluxes are external energy exchanges. Radiation should not be added to a closed-domain energy expectation as though it were an internal transfer. A held heating tendency and a freshly recomputed diagnostic flux can also refer to different states; comparing them without their cadence information can produce a misleading closure residual.

The combination of a boundary-layer scheme and resolved-grid SGS requires an explicit division of transport. Both can represent unresolved mixing, but they operate with different assumptions and length scales. Adding them together without a coupling policy can count the same transport twice. Nimbus's configuration admission and runtime owners restrict supported combinations; a paper describing either component independently does not settle the combined closure.

8.8 A numerical ledger can be exact while the physics is wrong

Let a completed step consist of states $X_0, X_1, \dots, X_m$, and let $E(X)$ be one consistently defined energy diagnostic. Then

$$E(X_m) - E(X_0) = \sum_{r=0}^{m-1} \left[E(X_{r+1}) - E(X_r) \right]$$

is a telescoping identity. It holds even if an operator implements an incorrect physical law. Recording endpoint differences establishes where the diagnostic changed, but does not independently establish why that change is physically justified.

A stronger report compares each change with a separately specified exchange,

$$R_r = E(X_{r+1}) - E(X_r) - \mathcal{W}_r - \mathcal{Q}_r - \mathcal{B}_r,$$

where $\mathcal{W}_r$ is the declared mechanical work, $\mathcal{Q}_r$ the thermal or phase exchange, and $\mathcal{B}_r$ the boundary contribution for that operation and energy basis. Those terms must be derived from the appropriate

physics and geometry. Defining $\mathcal{Q}_r$ to equal the measured endpoint difference by construction makes $R_r = 0$ without testing the thermal law.

Floating-point storage is another observable operation. A double-precision candidate cast back to a float field can change the diagnosed inventory. Nimbus distinguishes storage/rounding effects where required by an owner; those effects should not be relabeled as condensation, surface work or an unknown physical sink. Deterministic reduction can make the effect repeatable, but does not eliminate it.

The same discipline applies to conservation of water. A perfect domain sum can hide a spatially wrong transfer between cells; a correct cell source can coexist with a wrong boundary flux. Scientific validation therefore combines inventory closure, local reference states, geometric invariants and dynamic behavior. A single global residual is necessary for some claims and insufficient for all of them.

8.9 A worked control-volume example

Consider an illustrative flat cell with volume $100\,\mathrm{m} \times 100\,\mathrm{m} \times 50\,\mathrm{m} = 5 \times 10^5\,\mathrm{m}^3$ and dry density $1\,\mathrm{kg\,m^{-3}}$. Its dry mass is 5×10^5 kg. At $q_v = 0.01\,\mathrm{kg\,kg^{-1}}$, it contains 5000 kg of vapor. These numbers are chosen to expose the accounting; they are not a recommended grid or a Nimbus regression reference.

Suppose one face brings in 1000 kg of dry air carrying the same vapor mixing ratio while another removes 500 kg. The new dry mass is 500500 kg and vapor mass is $5000+10-5 = 5005$ kg. Dividing gives the same $q_v = 0.01$. Updating water with the entering and leaving fluxes but dividing by the old dry mass would instead produce a false concentration increase. Updating the mixing ratio before constructing its old inventory can create the opposite kind of error.

Now let local condensation transfer $\delta q = 2 \times 10^{-4}$ from vapor to cloud liquid at the transported endpoint. It transfers 100.1 kg of water:

$$W_v' = 5005 - 100.1, \qquad W_c' = W_c + 100.1.$$

Their sum is unchanged. With the illustrative held values $L_v = 2.5 \times 10^6\,\mathrm{J\,kg^{-1}}$ and $c_p = 1004\,\mathrm{J\,kg^{-1}\,K^{-1}}$, a constant-pressure estimate gives $\Delta T \simeq 0.498$ K. This estimate neglects temperature dependence, mixture heat capacity, saturation feedback and all transport during the source. Those are declared assumptions, not small corrections whose omission is universally justified.

If part of that liquid converts to rain, the atmospheric total water still need not change. If 20 kg subsequently falls through the physical bottom boundary, atmospheric water decreases by 20 kg and the surface precipitation inventory increases by 20 kg, subject to the chosen diagnostic units and area. Rain falling from an upper cell into this cell is an internal vertical exchange, not surface precipitation. The distinction depends on the physical interface owner, especially on a stretched or terrain-following column.

Finally, pressure projection and momentum evolution determine the new thermal and mechanical endpoint. The illustrative latent-heat estimate is not enough to predict that endpoint without the EOS and work assumptions. A complete step report would identify the carrier transport, phase transfer, sedimentation, pressure-related energy change, mixing, boundaries and storage effects. The example shows how to audit a step; it does not replace the actual source laws or their tests.

8.10 Publication should preserve the interpretation of the endpoint

After the numerical state is accepted, diagnostics reconstruct quantities such as temperature, vorticity, radar products and storm measures from that endpoint. A diagnostic can use a different spatial location from its inputs: for example, wind may be interpolated from faces to a scalar point for a plot. Its metadata must state that reconstruction. A visual field must not silently combine the new theta with an old pressure or a new geometry with an old domain origin.

The output layer publishes completion and geometry information so readers can identify a consistent frame. Compression and visual downsampling are additional representations of that frame. A block maximum is useful for retaining a small visible peak but is not a conservative volume average. A volume renderer's interpolated color is not a microphysics concentration. The architecture guide explains these later representations and the GPU path that produces pixels.

Restart has a stronger requirement than visualization. It must preserve the state needed by the next update, including private moments and relevant histories, and must reject incompatible continuation assumptions. A successful plot decode does not prove restart fidelity, and a successful restart does not by itself prove that the underlying physical experiment is valid.

8.11 Which assumptions travel with a result

A reproducible scientific result should state the physical domain and frame, scalar ownership and staggering, physical spacings and terrain geometry, base state and initial perturbations, transport and time-integration choices, boundary conditions, microphysics categories and moment units, closure and radiation choices, source cadences and required coefficient identities. It should also state precision, backend, threading and output transformations when those affect the comparison. These are the components of the problem actually solved.

The highest-level assumption is that a resolved, compressible grid model with bulk closures is an appropriate description for the question being asked. Subgrid distributions, unresolved turbulent motions and radiative optical properties are modeled rather than resolved particle by particle. Refining the grid does not remove every closure assumption. Conversely, matching one observed storm feature does not uniquely validate each closure: compensating errors can produce a similar visible outcome.

Verification asks whether the implemented operators satisfy their stated mathematical contracts. Validation asks whether the model is adequate for a physical use and whether its assumptions are supported by comparison with independent evidence. Long campaigns examine behavior beyond a bounded case; independent domain review examines the scientific interpretation. Nimbus's readiness terminology keeps those stages separate. Engineering-incomplete couplings remain work to be completed, even when a local formula, guard or reference test is already correct.

Chapter 9

Resolution, useful accuracy and the cost of an experiment

Nimbus's aim includes high-fidelity atmospheric experiments on ordinary workstations and MacBooks. Efficient algorithms and GPU acceleration can change which experiments are practical on a given machine. The mathematical question is what accuracy an experiment needs and which changes actually buy that accuracy. A smaller grid spacing, a faster timestep implementation and a more detailed rendered image are three different improvements. This chapter connects the physical question to discretization, computational cost and the evidence needed to increase resolution deliberately.

The discussion supplies planning models and experiment-design guidance. It does not assign a universal maximum grid to a laptop, recommend one spacing for all storms, or claim that Nimbus is already fully optimized. The architecture guide explains how to measure memory, data movement and execution, and what distributed GPU execution would require.

9.1 Define fidelity through a question and an observable

Let $X(t)$ denote the complete model state and $J[X]$ a quantity of interest. Examples include domain-integrated water, accumulated precipitation over a specified area and interval, an updraft-volume distribution, a cold-pool propagation speed, or a spectrum of resolved kinetic energy. Define the region, time interval, units and averaging operation before choosing the grid. A single maximum of vertical vorticity and an area-averaged circulation are not equivalent targets and need not converge at the same rate.

For example, a dry-mass-weighted mean mixing ratio over a fixed physical region $\mathcal{R}$ is

$$J_q(t) = \frac{\sum_{C \subset \mathcal{R}} M_C(t) q_C(t)}{\sum_{C \subset \mathcal{R}} M_C(t)}, \qquad M_C = \rho_{d,C} V_C.$$

This expression assumes the region is a union of owned physical cells. Partially intersected cells need an explicit overlap rule. Terrain volumes, moving domains and coarse/fine coverage change those weights; storage halos never contribute merely because they contain finite values. Comparing runs on different grids requires a common physical region rather than the same integer array slice.

High fidelity therefore has several components: the appropriate governing approximations, credible initial and boundary conditions, suitable closure assumptions, sufficient spatial and temporal resolution, and a numerical implementation that preserves the declared contracts. The objective is to make that combination affordable locally. Optimization can remove waste without changing the question; changing the question to fit the machine must be a deliberate scientific choice.

9.2 Grid spacing is not the smallest trustworthy feature

Sampling a wavelength at two points is a representability limit, not a promise that an evolution scheme propagates that wavelength accurately. To see the distinction, consider the teaching example of a centered derivative on a uniform collocated grid,

$$D_h f_j = \frac{f_{j+1} - f_{j-1}}{2h}, \qquad f_j = e^{ikjh}.$$

Substitution gives

$$D_h f_j = i\frac{\sin(kh)}{h} f_j, \qquad \frac{k_{\text{effective}}}{k} = \frac{\sin(kh)}{kh}.$$

At wavelengths $\lambda = 4h$ and $8h$, this ratio is approximately 0.637 and 0.900. At the two-cell alternating mode it is zero. The formula illustrates why a visibly sampled pattern can have a badly represented derivative. It is not the transfer function of Nimbus's full scheme: face staggering, reconstruction, upwinding, limiters, dissipation and time integration change the result. In particular, a face-centered gradient has a different symbol from this collocated example.

An effective-resolution study must examine the combined algorithm. Wave propagation and decay tests expose phase and amplitude errors; spectra and structure statistics expose where an evolving turbulent solution loses useful scale content. Skamarock's kinetic-energy-spectrum study is relevant methodological context. It does not establish Nimbus's effective resolution or justify transferring another model's cells-per-wavelength estimate without testing the Nimbus configuration.

The same issue matters in a vorticity rendering. Curl uses spatial derivatives of velocity, and its apparent small structures can be sensitive to filtering, stagger-to-center reconstruction and the derivative operator. A smooth GPU image can make a field easier to inspect while leaving its physical resolving power unchanged. Report the velocity source, curl definition, units, sampling, thresholds and display transformation with such a figure. The presence of fine visible structure is useful evidence to investigate, but its resolved scales must be established from the underlying fields and experiment.

9.3 Refinement changes the resolved and unresolved partition

In a filtered-flow interpretation, a representative subgrid stress is

$$\tau_{ij} = \overline{u_i u_j} - \overline{u}_i\,\overline{u}_j.$$

This expression states that filtering a product differs from multiplying filtered velocities. It is an illustrative incompressible form; a compressible closure may use density-weighted filtering and Nimbus's selected scheme has its own precise conventions. Refining the grid changes what is explicitly represented and what the closure must supply. It does not simply produce more samples of an otherwise unchanged turbulent model.

For an eddy-viscosity example, $K \sim (C\Delta)^2|S|$. Both the filter scale Δ and resolved strain $|S|$ can change on refinement. Holding the coefficient C fixed therefore does not hold the realized mixing fixed. The relationship between vertical and horizontal resolution also matters: a thin near-surface layer and a wide horizontal cell do not have the same numerical or closure behavior as an isotropic cell of equal volume.

Bryan, Wyngaard and Fritsch's study of deep moist convection motivates examining the scale assumptions of an LES closure alongside resolution. Its results are not a universal grid cutoff for every Nimbus experiment. Bulk microphysics, cloud optical properties, surface exchange and initial-condition uncertainty remain modeled even after important turbulent motions become resolved.

Refinement also needs compatible inputs. Preserve the physical size and amplitude of an initial perturbation rather than its radius in cells. Sample the same sounding and terrain with documented reconstruction. Keep sponge thickness and forcing regions in physical units where that defines the experiment. Recheck the realized base-state balance. Otherwise a purported resolution comparison changes the triggering or boundaries at the same time.

9.4 Why halving spacing can cost much more than eight times

For a fixed rectangular physical domain, ignoring halos and staggered end planes, the cell count is approximately

$$N \simeq \frac{L_x L_y L_z}{\Delta x\, \Delta y\, \Delta z}.$$

Refining each direction by factors (r_x, r_y, r_z) increases the cell count by $r_x r_y r_z$. The time cost also depends on the timestep. A schematic explicit wave/advection restriction is

$$\Delta t \left(\frac{|u| + c_x}{\Delta x} + \frac{|v| + c_y}{\Delta y} + \frac{|w| + c_z}{\Delta z} \right) \lesssim C_{\text{scheme}},$$

where the relevant characteristic speeds and stability constant depend on the operator and integrator. This is a dimensional planning relation, not Nimbus's exact timestep controller. Moving-grid transport uses velocity relative to the grid, and mapped coordinates require the corresponding metric factors. Split-explicit integration separates slow-step and acoustic restrictions; the cost can increase through more outer steps, more fast substeps, or both. Terrain metrics and local thin layers can determine the restrictive physical spacing.

If all spacings shrink by r, characteristic speeds stay comparable, the explicit timestep scales as $1/r$, and work per cell per step remains comparable, the same physical duration requires approximately r^4 work in three dimensions. Doubling linear resolution gives eight times the cells and about sixteen times the work under those assumptions. Horizontal-only refinement gives roughly r^3 work if the horizontal CFL restriction controls the step; if an unchanged vertical restriction already dominates, the initial increase can instead be closer to r^2. These are distinct cases, not conflicting rules.

For an explicit constant-diffusivity operator, a representative restriction is

$$K\Delta t \left(\Delta x^{-2} + \Delta y^{-2} + \Delta z^{-2}\right) \lesssim C_{\text{diff}}.$$

If this becomes controlling and K stays fixed, isotropic refinement can make the work scale as r^5. A grid-dependent SGS coefficient, implicit solve or subcycling changes that estimate. Positivity restrictions, sedimentation and microphysics can add costs that do not obey the simple wave model.

For example, increasing the number of cloudy cells can change the work distribution even when the total cell count scales regularly.

Illustrative change	State-size factor	Work factor for fixed duration
Halve all three spacings	8	About 16 if wave CFL controls
Halve horizontal spacings only	4	About 8 if horizontal CFL controls
Halve all three spacings, fixed explicit diffusivity controls	8	About 32
Double physical domain lengths at unchanged spacing	8	About 8 if step size and per-cell work stay comparable

The factors exclude changes in communication, cache behavior, output, initialization and iteration counts. They explain why optimization has large scientific value: removing repeated scans, avoidable synchronization or unneeded copies can make the next refinement practical without changing its equations. They also explain why a speedup on one grid cannot be extrapolated unchanged across a memory-capacity boundary. Actual complete-run measurements must determine the local hardware's practical range.

The same model quantifies an optimization opportunity. Suppose a complete matched workload becomes S times faster, and its refinement cost remains approximately proportional to h^{-4}. At a fixed time budget, that gain could support a linear refinement factor $r \simeq S^{1/4}$, provided memory and other constraints still fit. An illustrative sixteen-fold gain corresponds to halving all three spacings for about the original elapsed time. This is not a measured Nimbus speedup. The refined state still has eight times as many cells, so reducing unnecessary live storage can be as consequential as faster arithmetic. Alternatively, the same gain can support more realizations at an already adequate resolution. This is why optimization changes the accessible science on ordinary hardware rather than merely changing a benchmark number.

9.5 Memory and output belong in the same resolution calculation

With F simultaneously live scalar-equivalents and b bytes per scalar, a first payload estimate is bFN. It omits unequal field shapes, geometry, halos, stage duplication, private scheme state and output snapshots unless those have explicitly been included in F. Storage precision also differs from arithmetic precision. A float32 field with double-precision temporary work cannot be budgeted as though every live representation uses four bytes.

For a halo depth h around an owned local tile, the allocation ratio is

$$R_{\text{storage}} = \frac{(n_x + 2h)(n_y + 2h)(n_z + 2h)}{n_x n_y n_z}.$$

This simple rectangular example shows why small distributed tiles can have high overhead. Physical boundary layouts and C-grid fields change the exact count. Halos increase storage and communication, but not the physical atmospheric inventory used in a conservation check.

If F_o fields are saved for physical duration T at interval τ_o, an uncompressed payload estimate is

$$B_{\text{history}} \simeq bF_oN\left(1 + \lfloor T/\tau_o \rfloor\right).$$

It assumes an initial frame and fixed-size full fields, and excludes metadata, checkpoints and predictive dependencies. At fixed physical output cadence, isotropic refinement multiplies this payload by r^3. Saving every fixed number of shrinking timesteps can add another factor of r. A comparison should preserve the intended physical sampling rather than accidentally turning a resolution experiment into a different I/O workload.

Compression changes storage and transfer costs according to its actual mode and data. It does not make the uncompressed computational working set vanish. Some read paths can decode a bounded window; others still need a larger source working set. A measured compression ratio for one smooth field is not an admission guarantee for all fields or storm stages.

9.6 A refinement study compares the same mathematical experiment

For a smooth, sufficiently resolved verification problem, suppose $J(h) = J_* + Ch^p + o(h^p)$ with a stable leading error coefficient. For three grids $h, h/r, h/r^2$ with $r > 1$, an observed order is

$$p_{\text{obs}} = \frac{\log \left| \dfrac{J(h) - J(h/r)}{J(h/r) - J(h/r^2)} \right|}{\log r}.$$

If the differences are meaningful and the asymptotic model applies, the corresponding extrapolation is

$$J_* \approx J(h/r^2) + \frac{J(h/r^2) - J(h/r)}{r^{p_{\text{obs}}} - 1}.$$

These formulas are conditional estimates. Near-zero differences, nonmonotone behavior, phase shifts or changing physical closure can invalidate the interpretation. NASA's spatial convergence tutorial provides verification context. A formally high-order reconstruction does not establish the order of a complete moist, limited, split-coupled simulation.

Start with manufactured or analytical problems that isolate spatial and temporal errors. Refine time at a fixed grid, and refine space with temporal error controlled. Test hydrostatic balance, wave propagation, translated tracers, physical inventories and the selected source laws. For common-grid comparisons, restrict conserved inventories with their physical weights; arithmetic interpolation of mixing ratios is not generally conservative. Retain original fine-grid peaks as separate quantities if they are of interest.

Long nonlinear convective runs need additional interpretation. Small admissible changes can shift storm timing or location and eventually produce different realizations. Pointwise agreement at a late time is then a different question from agreement of precipitation, transport, spectra or event distributions. Compare those statistics on declared common scales and intervals. If a storm-relative alignment is used, report it alongside the unaligned comparison; it must not hide a propagation error. Neither visual similarity nor divergence alone settles numerical correctness.

9.7 Choose ensemble size and resolution together

Suppose an experiment estimates a mean diagnostic from m independent, identically distributed realizations with finite variance σ^2. Its sampling standard error scales as $\sigma/\sqrt{m}$. This does not describe discretization bias, model-form error or a single deterministic simulation. Snapshots from

one trajectory are usually correlated and must not be counted as independent realizations simply because many frames were saved.

A fixed compute budget can therefore be spent on finer individual runs, more realizations, longer sampling or a larger physical domain. Which choice helps depends on the quantity of interest and its dominant error. Reproducible perturbation seeds help paired comparisons, but do not create independence or validate the perturbation model. The local-hardware objective includes making ensembles practical, not only producing the finest isolated frame. Independent realizations are also a natural unit for spreading work across additional computers without domain-exchange traffic.

9.8 Keep numerical, physical and representation errors distinguishable

An experiment record should separate spatial and temporal discretization, closure and coefficient assumptions, initial and boundary uncertainty, sampling variability, floating-point effects, and saved-data transformations. Their effects interact; the list is not a theorem that their measured errors can always be added as independent scalar terms. It identifies what to vary and what to hold fixed in a controlled comparison.

For example, if saved velocity values have a pointwise error bound $|e_j| \leq \epsilon$, a centered derivative reconstructed from those saved values has representation error bounded by

$$\left|\frac{e_{j+1} - e_{j-1}}{2h}\right| \leq \frac{\epsilon}{h}.$$

A finer grid can amplify a fixed saved-field error in a derivative diagnostic. This bound assumes the pointwise premise and this specific stencil; it does not assert that every compression mode supplies that premise. Vorticity and other derivative products need their own error assessment. Lossy visualization output is also different from lossy prognostic state or restart, which can affect subsequent evolution.

The useful result of this process is a declared experiment envelope: the observable, physical setup, supported configuration, grid and timestep, input identities, output transformations, measured resource cost and observed sensitivity. That envelope can demonstrate demanding local experiments where the evidence supports them, and identify when additional optimization, memory, ensemble throughput or distributed execution is needed. Hardware placement follows the experiment and measurements; it is not a substitute for them.

Chapter 10

Solver source and scheme coverage

This index connects every top-level solver directory under `include/` and `src/` to the mathematical discussion. The architecture guide provides the corresponding module ownership, dataflow and execution contracts. Infrastructure, utilities and provisioning tools support these operators; they do not introduce additional atmospheric equations merely by existing in source.

Subsystem	Source and mathematical discussion
Core	interfaces, implementation; theory, conventions and limits.
Init	interfaces, implementation; theory, conventions and limits.
Soundings	interfaces, implementation; theory, conventions and limits.
Dynamics	interfaces, implementation; theory, conventions and limits.
Numerics	interfaces, implementation; theory, conventions and limits.
Boundary conditions	interfaces, implementation; theory, conventions and limits.
Microphysics	interfaces, implementation; theory, conventions and limits.
Radiation	interfaces, implementation; theory, conventions and limits.
Boundary layer	interfaces, implementation; theory, conventions and limits.
Turbulence	interfaces, implementation; theory, conventions and limits.
Terrain	interfaces, implementation; theory, conventions and limits.
Chaos	interfaces, implementation; theory, conventions and limits.
Radar	interfaces, implementation; theory, conventions and limits.
Diagnostics	interfaces, implementation; theory, conventions and limits.
Compute	interfaces, implementation; theory, conventions and limits.
Util	interfaces; theory, conventions and limits.
Tools	implementation; theory, conventions and limits.

10.1 Canonical scheme selectors

The following IDs are the canonical entries in the generated runtime catalog. Aliases and additional typed options are described by their configuration owners. A listed constructor does not establish that every grid, physics, output and backend combination is admitted.

Catalog family	Canonical IDs
Dynamics	`klemp_wilhelmson_split_cartesian_agrid`, `klemp_wilhelmson_split_cartesian_cgrid`
Advection	`tvd`, `weno5`, `none`
Diffusion	`explicit`, `implicit`
Time stepping	`rk3`, `rk4`, `split_explicit`
Microphysics	`kessler`, `lin`, `thompson`, `milbrandt`, `none`
Radiation	`simple_grey`, `rrtmg`, `none`
Boundary layer	`slab`, `ysu`, `mynn`, `none`
Turbulence	`smagorinsky`, `tke`, `none`
Chaos	`none`, `initial_conditions`, `boundary_layer`, `full_stochastic`
Terrain	`none`, `bell`, `schar`
Radar	`reflectivity`, `velocity`, `zdr`
Soundings	`auto`, `cm1`, `sharpy`

Initialization additionally separates file/parametric sounding sources, hodograph precedence and none/bubble/vortex triggers. Surface exchange and simple-grey optical-model selections are distinct typed choices within their owners. Those paths are explained in the initialization, radiation and mixing chapters rather than inferred from this catalog table.

The coverage check detects missing module or catalog entries and broken chapter references. It cannot prove that a derivation is correct, that a paper has been implemented completely or that a capability is qualified. Source review, mathematical checks, bounded tests and independent scientific review remain separate requirements.

Chapter 11

Scientific references and implementation provenance

Mathematics explains Nimbus's actual equations, assumptions and numerical representation. The references below supply scientific context; they do not imply that Nimbus implements every part of a publication or shares all of its discretization and thermodynamic choices. Current source, independent reference tests and configuration admission determine the implemented scope.

11.1 Dynamics and unresolved motion

Klemp and Wilhelmson (1978), *The Simulation of Three-Dimensional Convective Storm Dynamics*, Journal of the Atmospheric Sciences, 35, 1070–1096, supplies the compressible storm-model and acoustic-splitting background. Nimbus's dry rho–theta EOS, moisture-loading approximation, stage ordering and energy attribution must be read in the current dynamics code rather than inferred from that citation.

Moeng (1984), *A Large-Eddy-Simulation Model for the Study of Planetary Boundary-Layer Turbulence*, Journal of the Atmospheric Sciences, 41, 2052–2062, provides LES context. Nimbus's discrete stress, TKE, averaging, clipping and boundary-work contracts are specified in `src/turbulence` and its permanent tests; Nimbus does not thereby inherit the paper's complete spatial discretization.

Hong, Noh and Dudhia (2006), *A New Vertical Diffusion Package with an Explicit Treatment of Entrainment Processes*, Monthly Weather Review, 134, 2318–2341, describes the YSU nonlocal diffusion and explicit entrainment formulation. The source under `src/boundary_layer` defines the retained column equations, surface treatment and admitted Nimbus couplings.

11.2 Bulk microphysics

Lin, Farley and Orville (1983), *Bulk Parameterization of the Snow Field in a Cloud Model*, Journal of Climate and Applied Meteorology, 22, 1065–1092, is a primary mixed-phase bulk reference. Its wet-growth treatment and Nimbus's newer guarded material-temperature/vapor-mass extensions are explicitly distinguished in the mathematics guide.

Milbrandt and Yau (2005), *A Multimoment Bulk Microphysics Parameterization. Part II: A Proposed Three-Moment Closure and Scheme Description*, Journal of the Atmospheric Sciences, 62,

3065–3081, provides the spectral moment and process-closure background. Public triple-moment admission in Nimbus remains Engineering incomplete; a paper and an internal moment test do not complete its live source, energy, transport, restart and native-backend couplings.

Thompson, Field, Rasmussen and Hall (2008), *Explicit Forecasts of Winter Precipitation Using an Improved Bulk Microphysics Scheme. Part II: Implementation of a New Snow Parameterization*, Monthly Weather Review, 136, 5095–5115, describes the snow parameterization background, including its distribution and particle-shape assumptions. Nimbus's pinned Thompson implementation and offline table identities are specified by its maintained data guide. A scheme name alone is insufficient to identify a reproducible implementation.

11.3 Coefficient data and numerical contracts

RRTMG coefficient files are scientific data supplied by the AER model family; the native Nimbus implementation is C++, not a runtime call to reference Fortran. Keep coefficient identity and the official reference-column fixtures with the implementation. See the RRTMG guide and radiation tests.

For numerical changes, the permanent tests are the executable reference index: advection, diffusion, time stepping, EOS, and energy inventories. These checks supplement physical references rather than replacing convergence, observational validation or independent scientific review.

When adding a reference, identify the actual equation or assumption used and its Nimbus source/test owner. Avoid vague citations to an author's body of work, unsupported claims of identical schemes, or treating an archived campaign as a current runtime dependency. The repository readiness contract distinguishes Engineering incomplete, Ready, Campaign validated, Externally reviewed and Fully qualified; bibliography entries do not change those states.